

FlightRisk NYC

Predicting arrival delay before the aircraft moves

Datathon technical report — NYC Flights 2013

1. The problem, and why the usual version of it is too easy

An airline operations desk wants to know, at the moment a flight is scheduled to depart, whether it will land more than fifteen minutes late. Fifteen minutes is not an arbitrary cutoff: it is the US Bureau of Transportation Statistics definition of an on-time arrival, so it is the number airlines are measured on and the number that decides whether a connection holds.

The prediction is only worth anything if it arrives before the decisions do. Once the aircraft has pushed back, the crew is committed, the gate is reassigned, and the passenger with a 55-minute connection is already airborne. So we fix the decision point at the **scheduled departure time, before push-back**, and allow the model nothing that is determined after it.

That constraint is the whole project, because it is the one most published treatments of this dataset drop. `dep_delay` — how many minutes late the aircraft actually left — is sitting in the flights table, correlates with arrival delay at $\rho \approx 0.9$, and turns the problem into near-arithmetic. A model using it scores ROC-AUC above 0.90 and is operationally worthless, because by the time you know `dep_delay` you no longer need a prediction.

We did not simply exclude it. We trained both models and measured the gap, which turns out to be the most informative single number in the study:

Decision point	PR-AUC	ROC-AUC
Pre-push-back (deployable)	0.507	0.716
Post-push-back (adds <code>dep_delay</code>)	0.846	0.903

Two thirds of the apparent skill of a "flight delay model" is the observation that the plane left late. What remains — the 0.507 — is the part that is genuinely forecastable in advance, and it is the subject of this report.

1.1 Who this is for, and what it costs them

Delay is not a nuisance metric. The FAA's own commissioned study (Nextor) put the annual cost of US flight delay at **\$33 billion**, split roughly **\$16.7 billion onto passengers** — lost time, missed connections, unplanned meals and hotels — and **\$8.3 billion onto airlines** in crew, fuel and maintenance, with the balance falling on lost demand and downstream industries.¹ For a single minute the arithmetic is public and specific: **\$98.41 per block minute** of direct aircraft operating cost, filed by US passenger carriers on DOT Form 41 and published by Airlines for America — \$37.01 of crew, \$29.34 of fuel, \$18.35 of maintenance, \$9.76 of ownership.² Passenger time is valued at **\$47 per hour**, and that figure is not ours either: it is the FAA's own benefit-cost guidance.³

Three groups carry this, and they carry it unequally.

Operations desks are the ones who can act, and they are budget-constrained in a way that shapes the entire problem. A duty manager at a New York airport oversees several hundred departures a day and can meaningfully intervene on a few dozen — swap an airframe, hold a connection, pre-position a crew, move a gate. What they lack is not authority but a ranked list early enough to use. This is why the whole project is built around a fixed alert budget rather than a classification accuracy: the operational constraint is attention, not knowledge.

Connecting passengers carry the sharpest end. A delay of nine minutes is an inconvenience; a delay of sixty-five is a missed connection, a rebooking, and in the worst case an unplanned night. That is the reason this report spends \$5.5 on severity tiers rather than stopping at a binary label — the >60 minute tier is roughly the misconnection boundary, and the model reads it *better* than the 15-minute one (ROC-AUC 0.770 against 0.716).

Passengers to small destinations have the least slack and the least attention. A traveller to Los Angeles who misses a connection catches the next of many; a traveller to Burlington or Bangor may not fly again until tomorrow. This group is invisible in an accuracy metric and shows up only when you look at who a ranked alert list actually reaches, which is what §10 does.

Two things follow from taking those three groups seriously, and both changed the work rather than decorating it. First, the decision point had to be fixed *before* push-back, because that is when the desk still has options — which is what makes this problem hard and the published 0.90-AUC results irrelevant. Second, the model had to be evaluated as a rationing device rather than a classifier: §9 measures what a fixed budget reaches in minutes, passenger hours, dollars and CO₂, and §10 measures who it reaches.

2. Data

Source. NYC Flights 2013, available on Kaggle at <https://www.kaggle.com/datasets/aephidayatuloh/nyc-flights-2013> and canonically as the `nycflights13` R package (<https://github.com/tidyverse/nycflights13>). Five tables, all tracing back to US Bureau of Transportation Statistics on-time records, the FAA aircraft registry, and ASOS/NOAA hourly weather observations.

Table	Rows	Role
flights	336,776	one row per departure from EWR, JFK or LGA in 2013
weather	26,115	hourly observations at the three origin airports
planes	3,322	aircraft registry keyed on tail number
airports	1,458	coordinates, elevation, timezone
airlines	16	carrier code to name

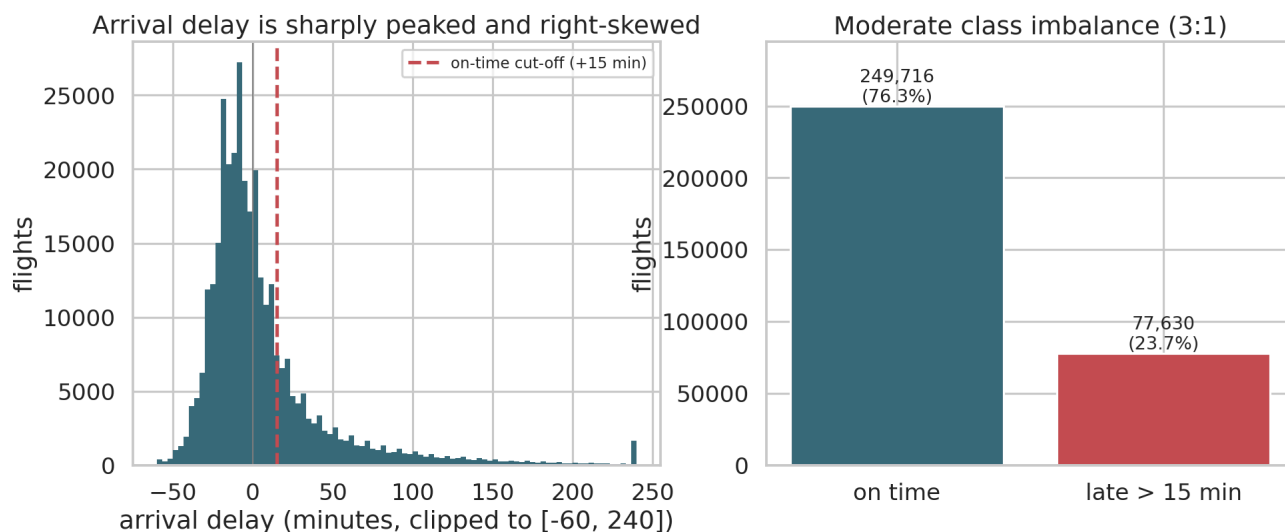
Labelling. 9,430 flights (2.8%) have no arrival delay because they were cancelled or diverted. They cannot carry an "arrived late" label and were removed, leaving **327,346 labelled flights**, of which **23.7% are late**.

This is a real limitation and we flag it rather than bury it. Cancellations are the most disruptive outcome of all, and they are not randomly distributed — they cluster in exactly the storm-and-holiday conditions the model is supposed to warn about. Dropping them understates operational disruption and makes the December test

period look milder than it was. A production system should model this as three outcomes (on time / late / cancelled), not two.

2.1 What the data looks like

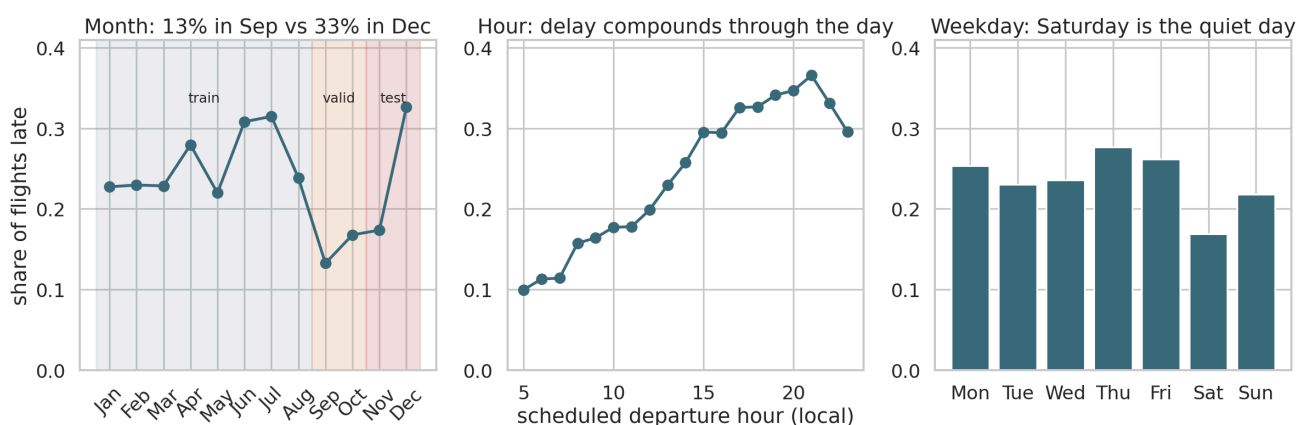
Target definition and balance



Arrival delay is sharply peaked and heavily right-skewed. The median flight arrives **5 minutes early**; the 90th percentile is **+52 minutes** and the 99th is **+190 minutes**. Because the distribution has so much mass just below the threshold, the 15-minute cutoff sits on a steep part of the curve — small shifts in conditions move a lot of flights across the line, which is precisely why the problem is learnable at all.

Class balance is a manageable 3:1. That is imbalanced enough that ROC-AUC flatters the model and accuracy is meaningless (a model predicting "on time" for everything scores 76%), so **PR-AUC is the headline metric throughout**.

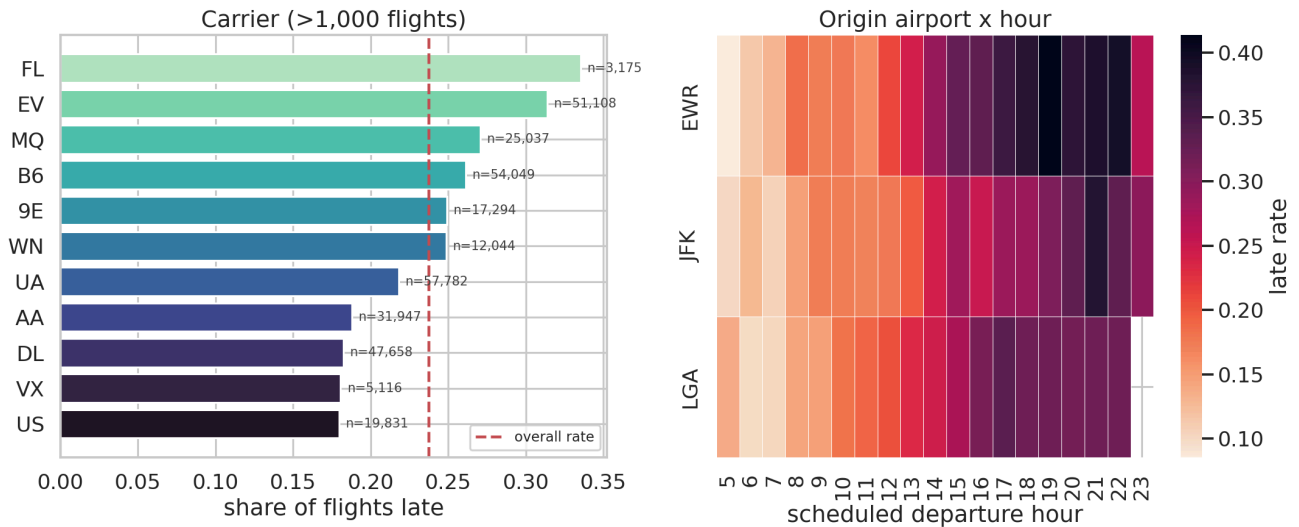
Systematic temporal structure in the delay rate



Three temporal structures, all strong:

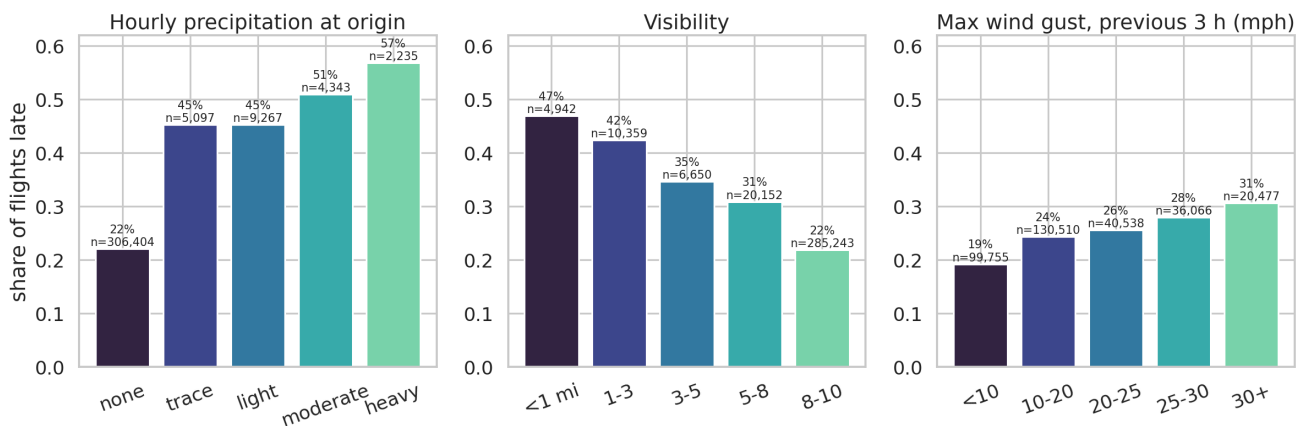
- **Month.** 13.2% late in September, 32.6% in December — a factor of 2.5. June and July are nearly as bad as December, for a different reason (thunderstorms rather than winter weather plus holiday load).
- **Hour.** 10% for the 05:00 departure bank, rising monotonically to 37% by 21:00. This is delay propagation made visible: the network starts each day clean and accumulates lateness.
- **Weekday.** Saturday is the quiet day at 17%; Thursday the worst at 27.5%.

Who and where: carrier and airport effects



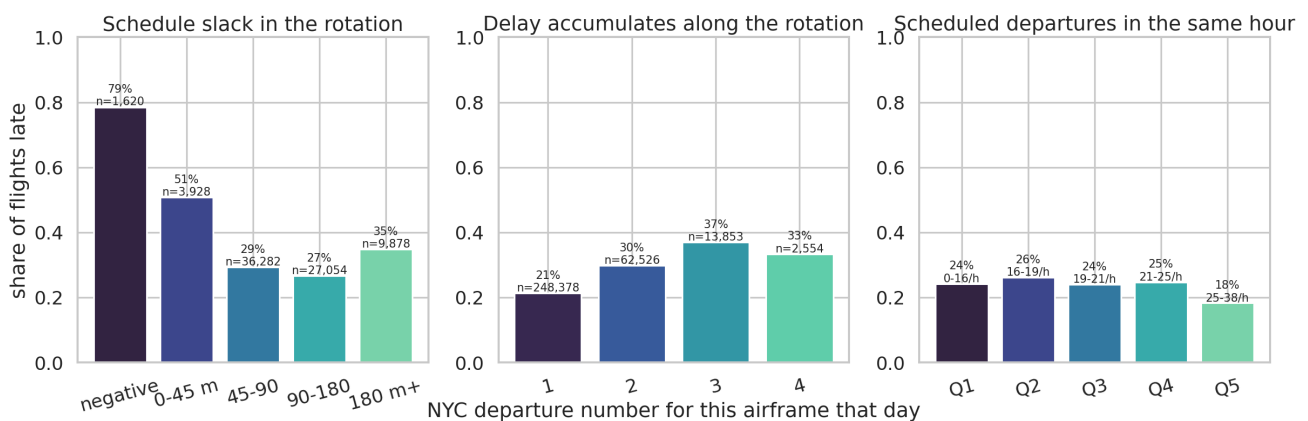
Carrier effects span a factor of three, from Hawaiian at 12.6% to Frontier at 37.3%, though the extremes are thin-volume operators. Among the large carriers the spread is still wide: Delta 18.2% against ExpressJet 31.4%. Newark is the worst of the three airports (25.6%) and LaGuardia the best (22.4%), but the airport effect is small next to the hour-of-day effect.

Weather at the NYC origin moves the delay rate by 20+ points



Weather at the origin is the strongest single driver visible in the raw data. Dry hours run 22.1% late; hours with any precipitation run **47.7%**. Heavy precipitation reaches 57%. Visibility below 3 miles doubles the rate from 22.9% to 45.5%. Wind gusts matter but less dramatically — 19% below 10 mph, 31% above 30 mph.

Operational structure: turnaround, rotation position, congestion



The operational panel is where the feature engineering earns its keep, and it also contains a mistake we made and corrected — see §3.3.

2.2 Data quality issues found and handled

- **Impossible wind speeds.** The ASOS feed contains a 1,048 mph reading and several other physically impossible values. Clipped at 60 mph (gusts at 90) rather than dropped, so the hour survives for its other variables.
- **Missing wind gusts.** Recorded only when gusting; filled with the sustained wind speed, which is the correct physical default.
- **Unmatched weather hours.** 0.47% of flights have no observation at their scheduled hour. Filled with the origin-and-month median.
- **Unmatched aircraft.** 16.3% of tail numbers are absent from the FAA registry. Left as missing rather than imputed — LightGBM learns a missing-value direction natively, and "this airframe is not in the registry" is itself weakly informative.
- **A non-unique join key, found late.** The clocks went back on 3 November 2013, so local hour 01:00 occurs twice at each airport and the weather table contains six rows whose `(origin, date, hour)` key is not unique. A left join on a non-unique key silently duplicates flights. This is invisible in the main model — nothing departs at 01:00 — and only surfaced when the forecast-horizon experiment (§5.6) shifted the join backwards in time and mapped real departures onto that hour, tripping an assertion on the row count. Fixed by keeping the first observation of the repeated hour; two tests now guard it. The lesson is that the bug was latent and harmless for months of work, and would have stayed latent without an experiment that stressed the join in a new direction.
- **Timezone correctness.** Scheduled block time cannot be computed by subtracting two clock times, because arrival is in the destination's timezone. `tests/test_features.py` verifies that the corrected block time exceeds actual air time by a median of 31 minutes (plausible taxi) and that JFK–LAX comes out near six hours rather than three or nine.

3. Feature engineering

67 features in the deployable configuration, in seven families. The organising question for every one of them was: *does this quantity exist, in this form, at the scheduled departure time?*

3.1 The three kinds of information, and which are legitimate

Published-schedule facts are known months ahead and cannot leak. Congestion counts fall here: how many flights are *scheduled* to leave Newark in this hour, how many share this 15-minute slot, what share of the slot belongs to this carrier. We compute these over the whole calendar year, including the test period, and that is correct — the timetable for December was published in advance and contains no information about December outcomes.

Contemporaneous observations exist at the decision point. Origin weather at the scheduled hour qualifies, along with rolling 3- and 6-hour precipitation totals, minimum visibility, maximum gust and pressure tendency, all of which look strictly backwards.

Outcome-derived statistics must be fitted on the training period only. Eight smoothed historical late rates (by carrier, destination, route, tail number, origin × hour, carrier × destination, destination × hour, carrier × hour) use Bayesian shrinkage toward the global prior with $m = 50$ pseudo-flights, so a route flown twenty times does not get a confident rate.

Inside the training period these encodings are computed **out-of-fold** across five folds. This is not a formality. `tailnum` has roughly 4,000 levels; an in-fold encoding would let each flight's own label into its own feature and the model would memorise the training set. The test suite asserts that `te_tailnum` alone reaches an AUC between 0.50 and 0.62 — a leaked version would approach 1.0.

3.2 A target encoding we had to throw away

Our first encoding set included a `destination × month` key. It looked excellent in training. It is worthless at deployment, and the reason is structural: with a temporal split, **no test month ever appears in the training period**, so every test row falls back to the global prior. A feature that is informative in training and constant in production is not merely useless, it actively displaces capacity the model could have spent on something that generalises.

The general rule we adopted: **any encoding key must be recurrent across the split boundary**. `dest × hour` is fine because December has the same 24 hours as January. `dest × month` is not. A test in the suite now enforces this by asserting that fewer than half of test rows fall back to the prior for any encoding.

3.3 Aircraft rotation: a feature we got wrong first

Delay propagation is the dominant mechanism in airline operations — a late inbound aircraft makes the next departure late — so we wanted a turnaround feature. The first attempt computed, per tail number, the gap between the previous leg's scheduled arrival and this departure.

The EDA figure exposed the error immediately: flights with a "turnaround" under 45 minutes were 81% late, but there were only 186 of them, and several had *negative* turnarounds. The bug is conceptual. **This dataset records only flights departing NYC**. An aircraft that leaves JFK for Miami must fly back to New York before it can leave again, and that return leg is invisible. The gap we measured spans an entire unobserved round trip, so a "45-minute turnaround" was a data artefact, not a tight turn.

The fix subtracts the previous leg's scheduled block time as an estimate of the unobserved return, leaving the genuine schedule slack:

```
rotation_slack_min = (sched_dep - prev_sched_arr) - prev_sched_block
```

The corrected feature behaves like a real operational quantity:

Schedule slack	Late rate	n
negative (impossible timetable)	79%	1,620
0–45 min	51%	3,928
45–90 min	29%	36,282
90–180 min	27%	27,054
180 min+	35%	9,878

Monotone across the operationally meaningful range, with a sensible uptick at the long tail (very long gaps mean the airframe sat idle through a disrupted day). We kept the mistake in the report because the diagnosis — *the data generating process does not contain what I assumed it did* — is the lesson.

The same reasoning constrains the post-push-back mode: the inbound arrival delay is only used when the previous leg is same-day **and** its actual arrival preceded our scheduled departure. A test verifies this holds for every flagged row.

3.4 The seven families

Family	n	Examples
Weather	17	precipitation and 3/6-hour totals, visibility and 3-hour minimum, wind gust and 3-hour maximum, pressure tendency, temperature, humidity
Historical rates	8	smoothed out-of-fold late rates by carrier, route, airframe, origin × hour
Calendar	12	month, day of week, day of year, holiday window, departure hour (raw and cyclical), red-eye flag
Route & schedule	11	distance, scheduled block time, block time versus route median, implied speed, destination coordinates and elevation
Congestion	7	departures from the origin in the hour and 15-minute slot, carrier share of the slot, system-wide NYC departures
Rotation	5	schedule slack, gap since previous NYC leg, tight-turn flag, leg number of the day
Aircraft	6	age, seats, engines, engine type, manufacturer, airframe class

4. Experimental design

4.1 The split is temporal, and deliberately unkind

Train **Jan–Aug** (217,727 flights), validate **Sep–Oct** (55,628), test **Nov–Dec** (53,991). Never random.

A random split would let the model see flights that happen *after* the ones it is scored on, and with congestion and weather shared across flights in the same hour, neighbouring rows are close to duplicates. Random-split scores on this dataset are inflated and not comparable to ours.

The temporal split is uncomfortable by construction, and we chose it anyway:

Period	Late rate
Train (Jan–Aug)	25.6%
Validation (Sep–Oct)	15.1%
Test (Nov–Dec)	25.0%

Validation sits at a base rate ten points below both neighbours, because September and October 2013 were unusually smooth. This is exactly the situation a deployed model faces — you tune on the recent past and deploy into a future that does not resemble it — and it produces the most interesting result in the study (§6).

4.2 Hyperparameter search

40 random draws over nine LightGBM hyperparameters, each scored by PR-AUC on a **forward-chaining three-fold time-series split inside the training period**: fold k trains on the earliest months and scores the block that follows, never the reverse. Early stopping at 100 rounds within each fold.

Random search rather than grid: with nine interacting dimensions, a grid of 40 points would pin each dimension to one or two values, whereas 40 random draws give 40 distinct values per dimension.

The validation block was deliberately excluded from the search and reserved for early stopping, calibration and threshold selection, so that **the test set was scored exactly once, at the end**.

Search spread was narrow — best 0.5376, median 0.5328, worst 0.5258 across 40 draws. The problem is not hyperparameter-sensitive, which is itself worth knowing: effort spent on features paid far better than effort spent on tuning.

Selected configuration:

```
num_leaves 191 · max_depth 8 · learning_rate 0.02 · min_child_samples 100  
feature_fraction 0.7 · bagging_fraction 0.9 · lambda_l1 5.0 · lambda_l2 2.0  
min_split_gain 0.05 · 391 trees at early stopping
```

Fold scores 0.490 / 0.560 / 0.563 — the first fold is meaningfully harder, reflecting how much less history it trains on.

4.3 Baselines

Four, each answering a different objection:

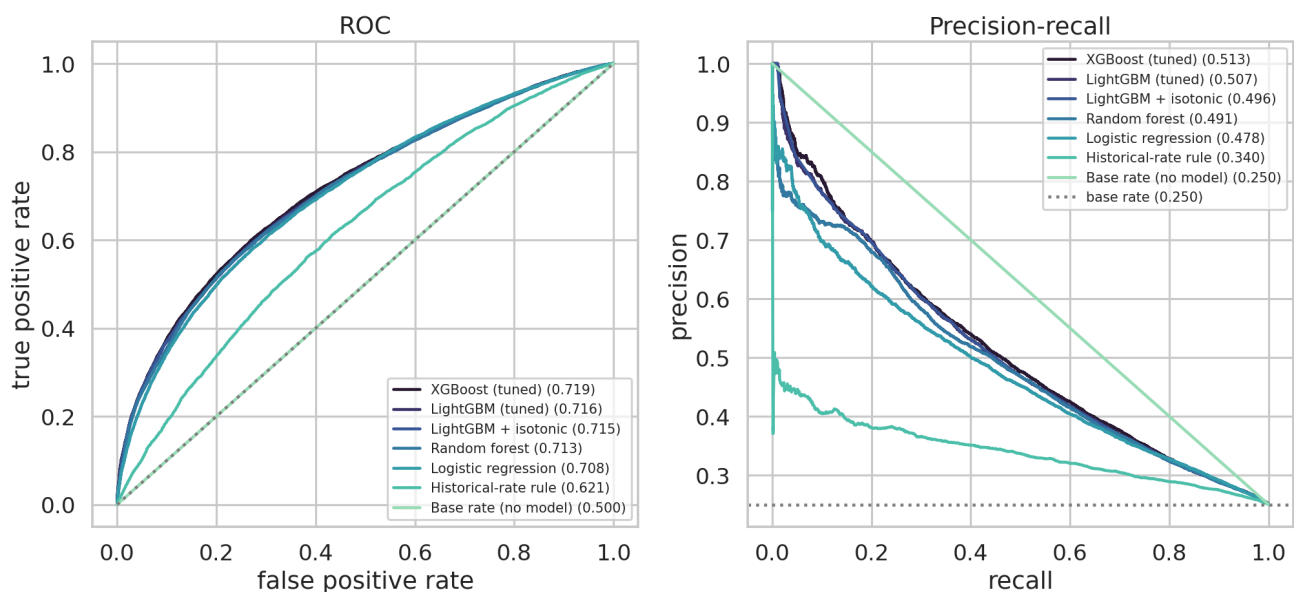
1. **Base rate** — predicts the training prior for everything. ROC-AUC 0.500 by construction; the floor.
2. **Historical-rate rule** — what an operations analyst writes without machine learning: look up how often this carrier flew this route at this hour and was late. Built from the same smoothed encodings the model gets, so it is a fair comparison rather than a strawman.
3. **Logistic regression** — median imputation, standardisation, one-hot with rare levels pooled. Tests whether the problem needs non-linearity.
4. **Random forest** — capped at 150 trees, depth 14. Tests whether it needs *boosting* specifically.

5. Results

All numbers on the untouched Nov–Dec test period: 53,991 flights, 25.0% late.

Model	PR-AUC	ROC-AUC	Brier	Log loss
Base rate	0.250	0.500	0.188	0.562
Historical-rate rule	0.340	0.621	0.182	0.547
Logistic regression	0.478	0.708	0.166	0.509
Random forest	0.491	0.713	0.164	0.504
LightGBM (tuned)	0.507	0.716	0.167	0.519
XGBoost (tuned)	0.513	0.719	0.168	0.518
<i>LightGBM, post-push-back</i>	<i>0.846</i>	<i>0.903</i>	<i>0.097</i>	<i>0.327</i>

Held-out test period (Nov-Dec 2013), pre-flight features only

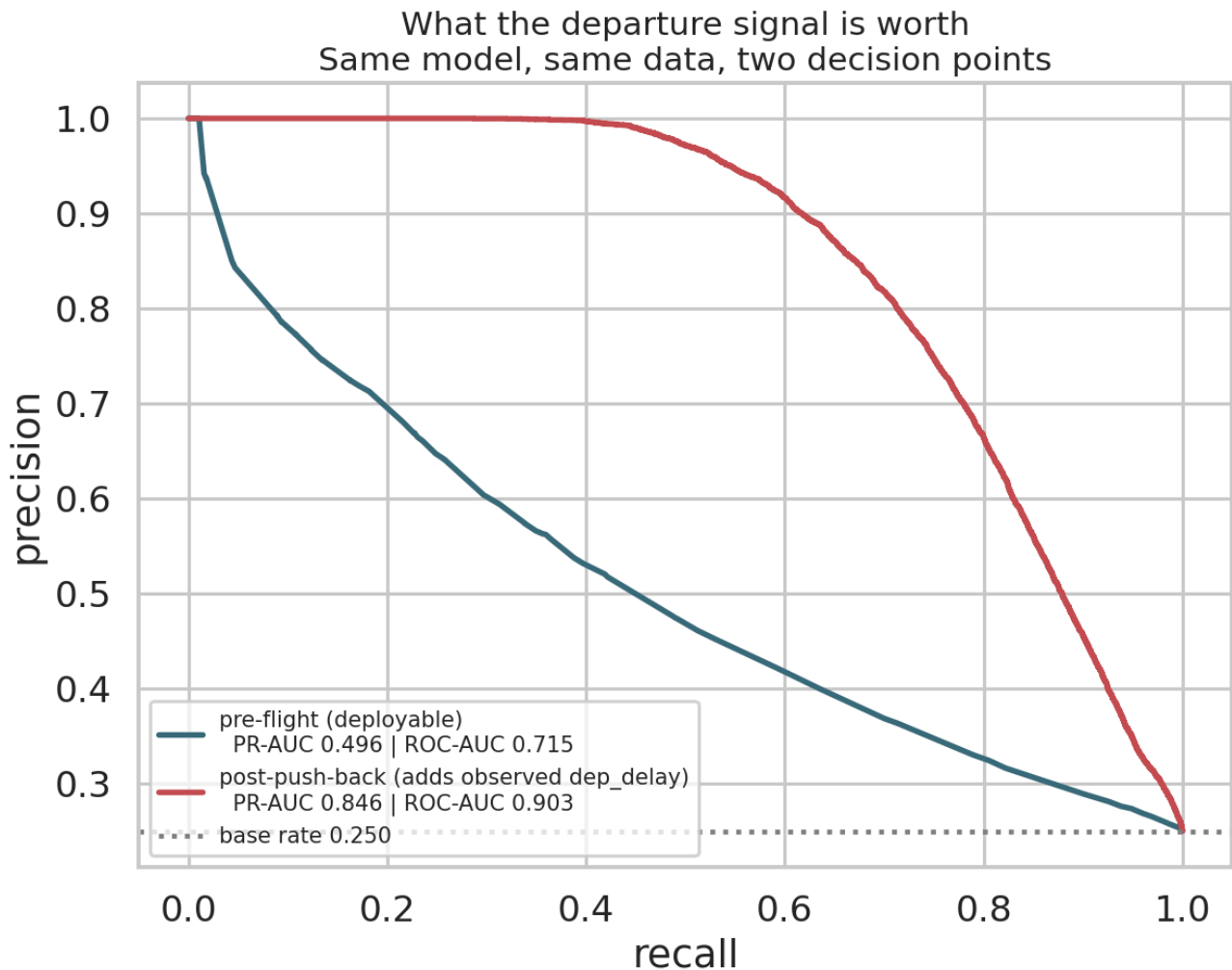


Reading these honestly:

- The tuned model doubles PR-AUC over the base rate (0.507 against 0.250) and beats the no-machine-learning rule by half again (0.340).
- **The gain from gradient boosting over logistic regression is 0.029 PR-AUC.** Real, but small next to the 0.138 that logistic regression itself gains over the historical-rate rule. Most of the value came from constructing the features, not from the model class.
- **XGBoost edges LightGBM, and the first version of this comparison was unfair.** Initially XGBoost got 8 search draws to LightGBM's 40, and its categorical features were one-hot expanded while LightGBM used native categorical splits. On that footing the two looked identical (0.508 vs 0.507). Rerun with 40 draws over a matched nine-dimensional space, `grow_policy="lossguide"` so both grow leaf-wise, and `enable_categorical` so both see the same representation, XGBoost is ahead on cross-validation at every percentile (median draw 0.5371 against 0.5328) and on the test set (0.513 against 0.507). The gap is small, but the lesson is not: **a library comparison is mostly a comparison of the search budget and the feature representation each library was given.** Ours was accidentally rigged, and it took deliberately re-leveling it to notice.
- Random forest has the best Brier score and log loss despite worse ranking, because averaging bootstrap votes produces conservative, well-spread probabilities. Discrimination and calibration are separate

properties, and §6 makes that separation the point.

5.1 The prediction-horizon experiment



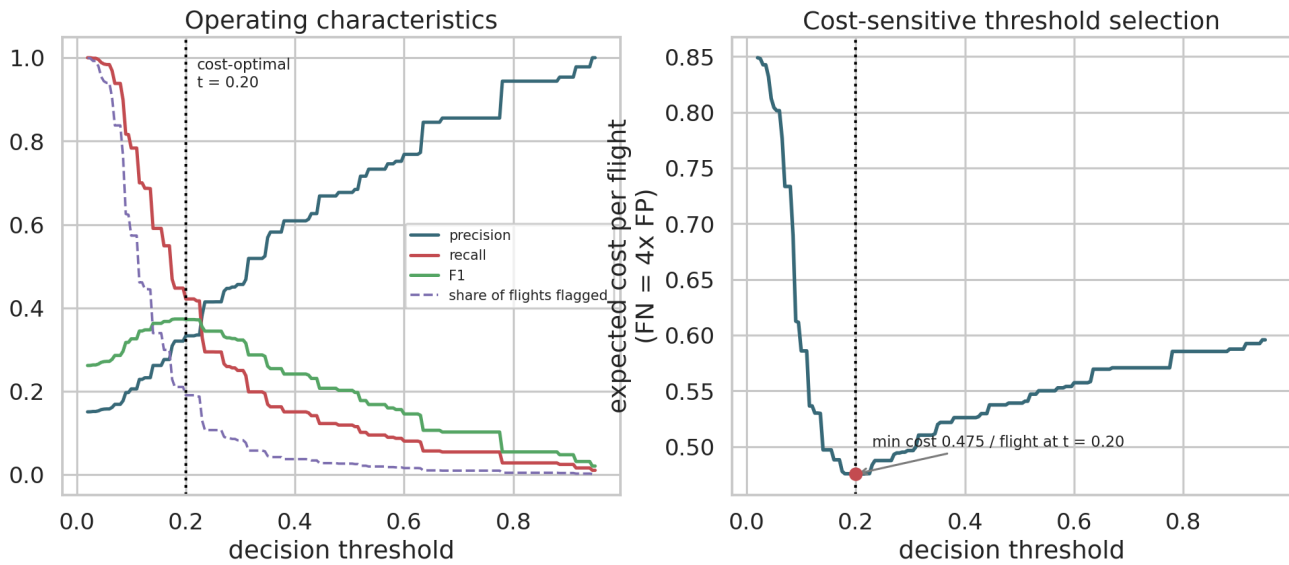
Same model, same features, same split — the only change is admitting `dep_delay` and the inbound leg's actual delay. PR-AUC goes from 0.507 to 0.846; ROC-AUC from 0.716 to 0.903.

The interpretation is not that our model is weak. It is that **arrival delay is mostly departure delay plus noise**, and departure delay is not knowable in advance. Any published result in the 0.90 ROC-AUC range on this dataset is answering the post-push-back question. The two numbers are not comparable and should never be quoted side by side without the horizon attached.

5.2 Turning probabilities into decisions

A probability is not a decision. We define an explicit cost model: a false alarm wastes agent time and erodes trust; a missed delay causes misconnections, rebooking and compensation. We set the missed-delay cost at **4× the false alarm** — a policy choice, stated openly and swept rather than hidden.

The 0.5 default is the wrong threshold for this decision

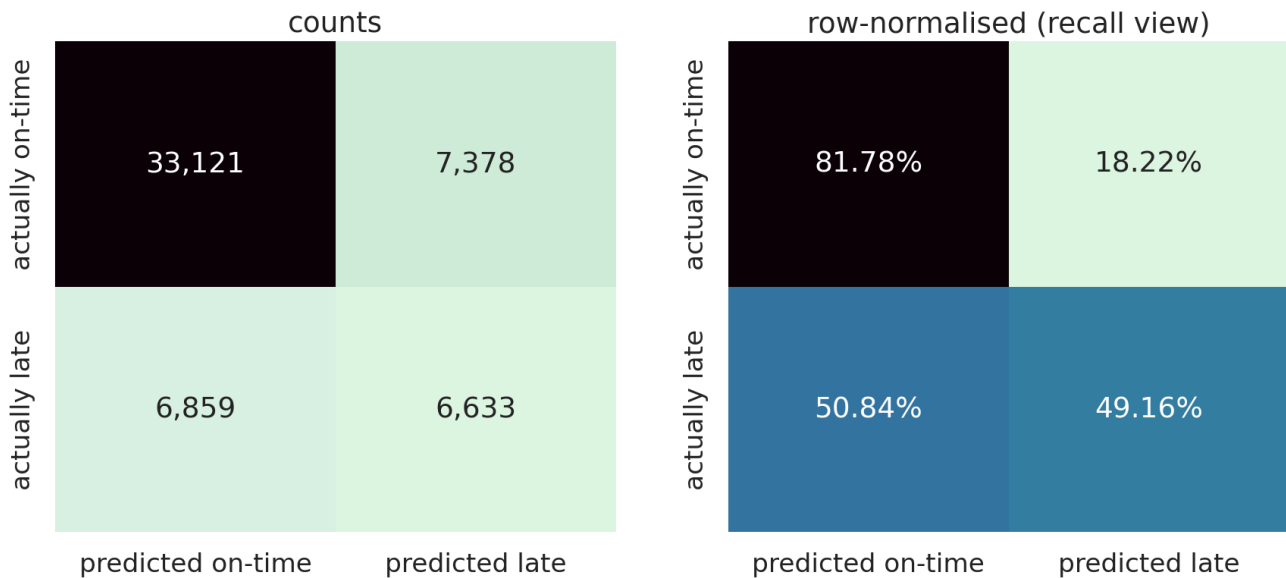


Minimising expected cost on the validation period gives a threshold of **0.20**, not 0.5. On the test period that yields:

Precision	47.3%
Recall	49.2%
F1	0.482
Share of flights flagged	26.0%
Expected cost per flight	0.645

Against the alternatives: alerting on nothing costs 1.000 per flight and alerting on *everything* costs 0.750. The cost-optimal threshold beats the better of those by 14%. The default 0.5 threshold costs **0.844** — 31% worse than the tuned threshold, and worse than the trivial policy of alerting on every single flight. **A well-fitted model deployed at the default threshold would have been actively counter-productive here.** Choosing the threshold by cost rather than convention is the difference between a model that helps and one that does not.

Confusion matrix at the cost-optimal threshold $t = 0.20$ (test period)



5.3 The operational view

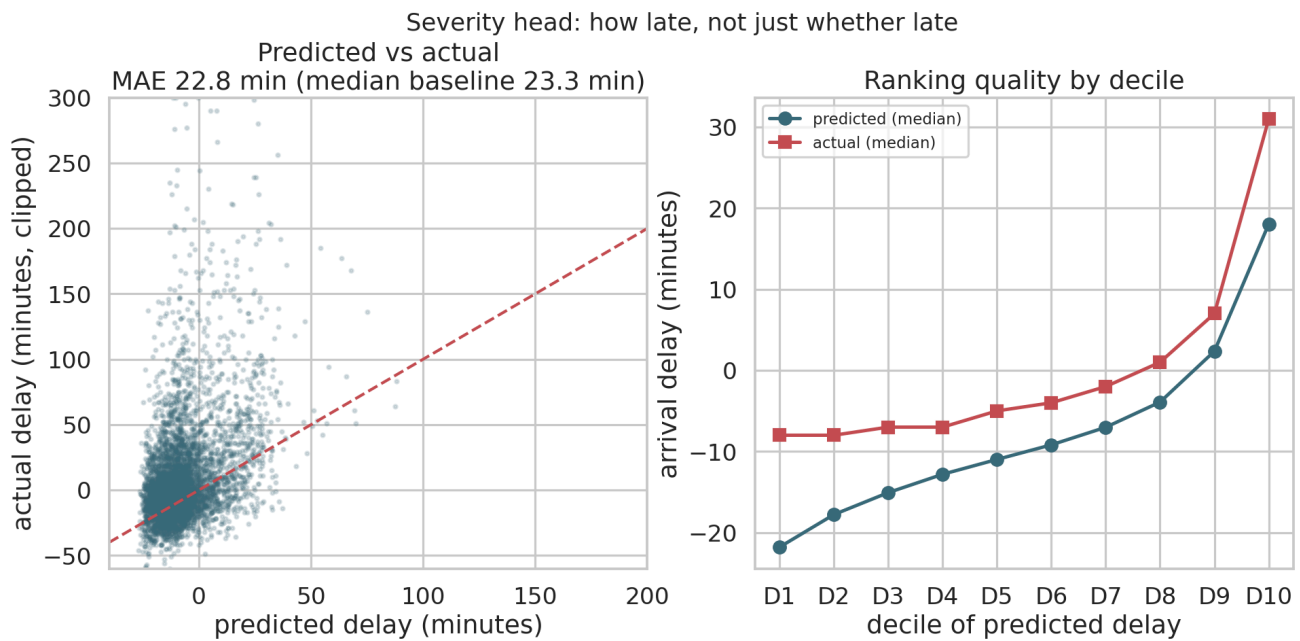
A desk cannot act on 26% of flights. Fixing the alerting budget at the top 10% by predicted risk:

Flights flagged	5,399
Of which actually late	64.1%
Share of all late flights caught	25.7%
Lift over base rate	2.56x

Two out of every three flights the model puts at the top of the list do arrive late, against a background rate of one in four. For a triage tool this is the number that matters, and it is considerably more encouraging than ROC-AUC 0.716 sounds.

5.4 Severity: how late, not just whether late

A second LightGBM regressor predicts arrival delay in minutes, trained with an L1 objective because the right tail would otherwise dominate a squared-error fit.



The honest summary is mixed, and the two halves point in opposite directions:

- **Point accuracy is barely better than a constant.** MAE 22.8 minutes against 23.3 for always predicting the training median — a 2.2% improvement. Three quarters of flights are on time and L1 loss rewards predicting the median, so it does.
- **Ranking is genuinely useful.** Spearman $\rho = 0.332$, and the decile medians rise monotonically from -8 minutes in the lowest decile to **+31 minutes in the highest**, where 64% of flights are late. On late flights specifically, MAE improves from 60.4 to 57.0 minutes.

So the severity head should be presented as a *sorting* tool, not a *forecasting* tool. "This flight is in the worst decile, where the typical delay is half an hour" is supportable. "This flight will be 31 minutes late" is not.

That verdict was weak enough to be worth a second attempt, and the second attempt is in §5.6.

5.5 Severity, reframed: tiers and quantiles

The conditional-mean regressor answers a question nobody asks — "what is the expected delay of a randomly chosen flight" — and L1 loss on a distribution that is 76% on-time is minimised by predicting the median, so it does. Three replacements, each matched to a decision that gets made.

Severity tiers. Separate classifiers for delay > 15 , > 60 and > 120 minutes. The thresholds are operationally distinct: 15 minutes is the on-time metric, 60 breaks most connections, 120 enters compensation territory.

Tier	Base rate	PR-AUC	ROC-AUC	Precision in riskiest 10%	Lift
> 15 min	25.0%	0.505	0.716	63.7%	2.55×
> 60 min	7.4%	0.293	0.770	30.5%	4.10×
> 120 min	2.3%	0.168	0.793	11.9%	5.14×

Severe delays are more predictable than marginal ones, not less. ROC-AUC rises from 0.716 to 0.793 as the tier gets rarer, and lift more than doubles. That is the opposite of the usual rare-event story, and it makes sense mechanically: a two-hour delay needs a real cause — a storm, a closed runway, a badly broken

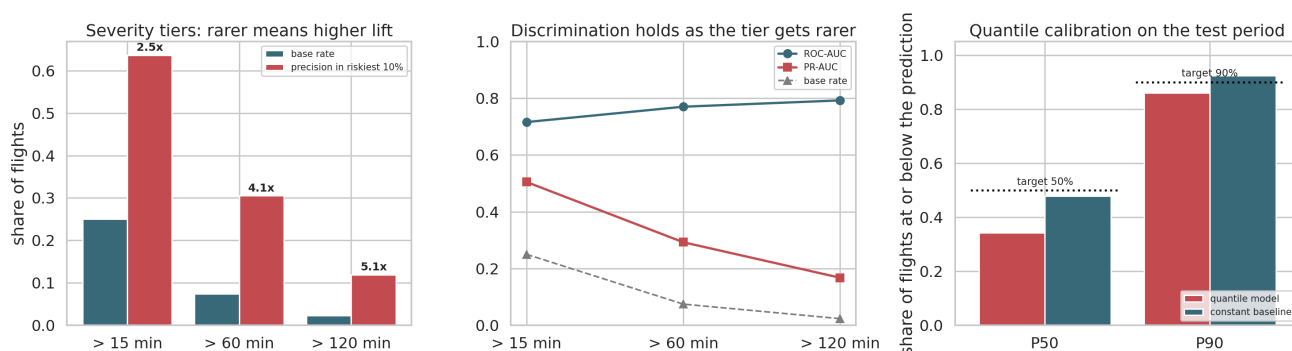
rotation — and those causes are in the feature set. A 20-minute delay is mostly noise. The model is best exactly where the consequences are worst.

Quantile heads. Quantile regression at the 50th and 90th percentiles, scored with pinball loss against a constant-quantile baseline.

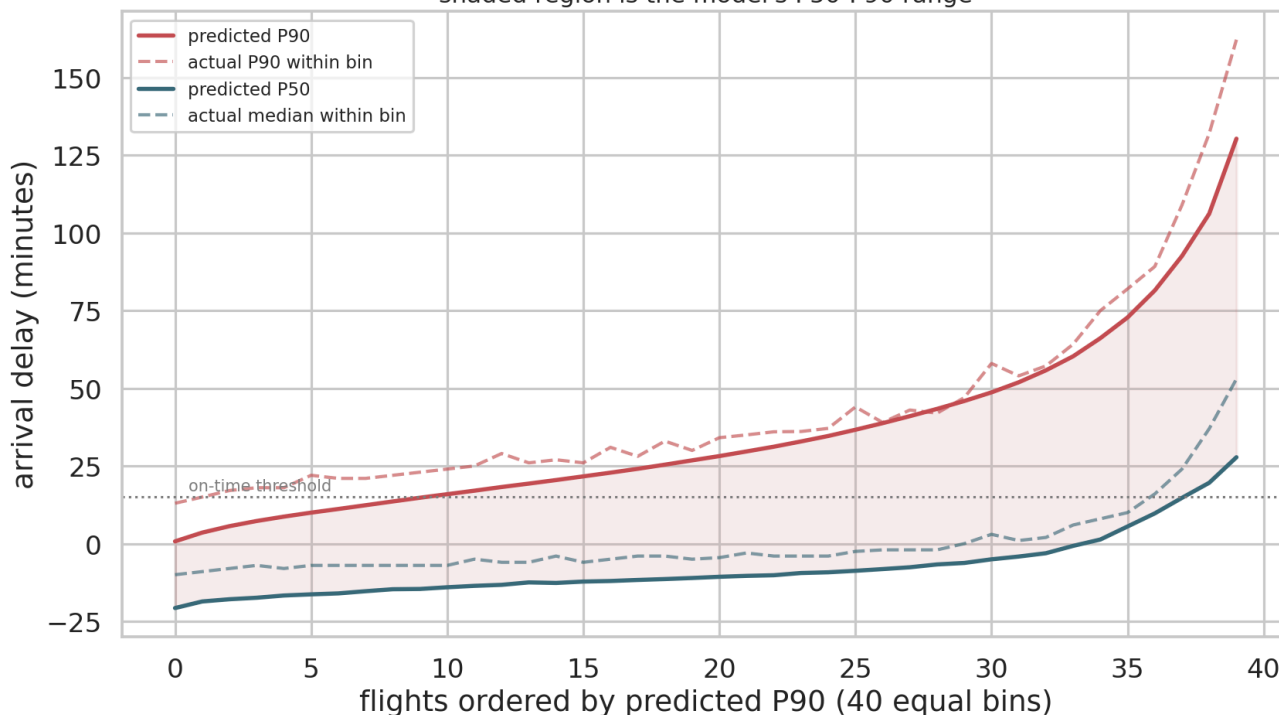
Head	Pinball	Constant baseline	Improvement	Coverage (target)
P50	11.395	11.631	2.0%	0.343 (0.50)
P90	7.757	9.281	16.4%	0.860 (0.90)

Same split as the tiers: the **upper tail is predictable and the centre is not**. The P90 head is 16.4% better than a constant, the P50 head is 2.0% better, i.e. nothing. Both under-cover on the test period (0.860 against a 0.90 target) for the same December reason as \$6 — the quantile heads inherit the level shift, and would need the same rolling correction.

Severity, reframed: tiers and quantiles instead of a conditional mean



The P90 band tracks the real upper tail
shaded region is the model's P50-P90 range



Conditional on lateness. Trained only on flights that did arrive late, MAE improves from 32.1 to 29.5 minutes, a 7.9% gain — three and a half times the 2.2% the unconditional model managed, because the target

no longer has a 76%-on-time spike at its centre.

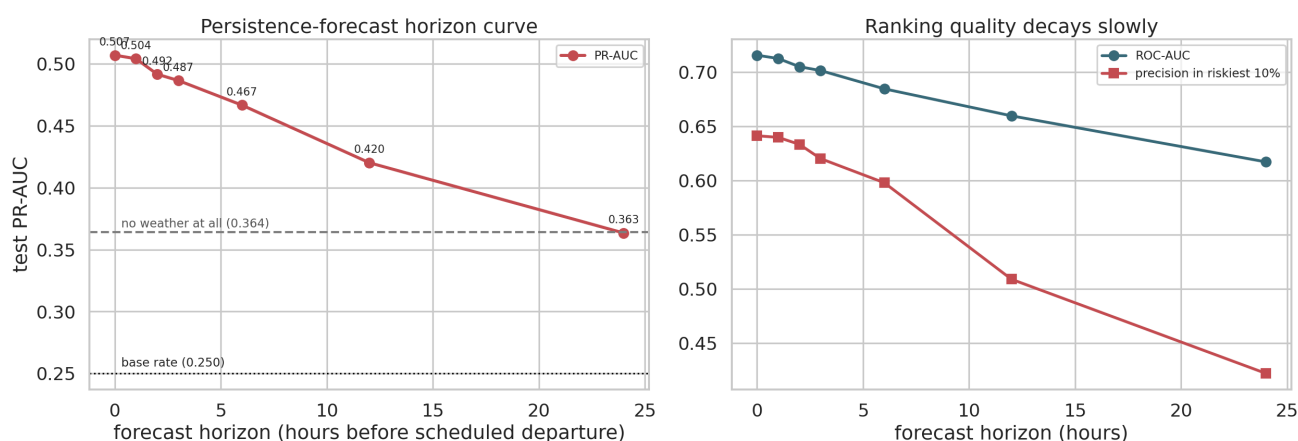
The reframing is the finding: the same features that barely support a conditional mean comfortably support a tail estimate and a tier classifier. Choosing the wrong output format made a usable model look useless.

5.6 How far ahead can this actually be predicted?

The model uses the weather observation at the scheduled departure hour. That is legitimate at a zero-hour horizon, but the ablation (§7.1) showed weather carries 0.141 of the 0.507, so "you would need a forecast to go further ahead" cannot be left as an unquantified caveat.

For a horizon of h hours each flight was given the observation from h hours before its scheduled departure, and the model retrained from scratch. That is a **persistence forecast** — the crudest one possible, "conditions will be what they are now". Real numerical weather prediction beats persistence at every horizon past an hour or two, so this curve is a **lower bound** on what a forecast-fed model would achieve.

Weather at h hours' notice: a lower bound on forecast-fed performance



Horizon	PR-AUC	ROC-AUC	Precision in riskiest 10%	Share of weather's contribution retained
0 h	0.507	0.716	64.1%	100%
1 h	0.504	0.713	64.0%	98%
2 h	0.492	0.705	63.3%	89%
3 h	0.487	0.701	62.0%	86%
6 h	0.467	0.685	59.8%	72%
12 h	0.420	0.660	50.9%	39%
24 h	0.363	0.617	42.2%	0%

Three things to take from it:

1. **A three-hour planning horizon costs almost nothing** — 0.020 PR-AUC, 86% of weather's contribution retained, and that is with the worst possible forecast. Six hours still retains 72%. The system is deployable well before push-back, which was the open question.
2. **Twenty-four hours is worthless, exactly.** The $h=24$ model scores 0.3634; the model trained with no weather at all scores 0.3640. Yesterday's weather at this hour carries no usable information about today's.

That the curve lands precisely on the no-weather floor is a built-in sanity check that the experiment measures what it claims to.

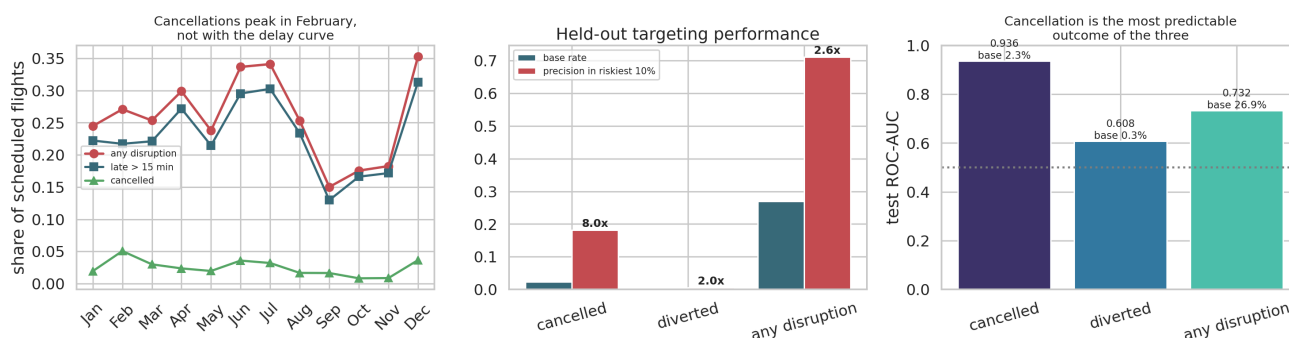
- Ranking decays more slowly than calibration-sensitive metrics.** Precision in the riskiest 10% holds above 60% out to three hours. Even at 12 hours the top decile is half late, against a 25% base rate.

5.7 Recovering the flights that were thrown away

§2 dropped 9,430 cancelled and diverted flights because they have no arrival delay, and flagged it as a limitation. It is the worst kind of limitation: cancellation is the most disruptive outcome for a passenger, and cancellations are not randomly distributed — 5.1% of February departures against 0.8% in October. Dropping them makes the year look calmer than it was, precisely in the weeks the model exists to warn about.

So keep all 336,776 flights and model three outcomes on the identical pre-flight feature set. `is_cancelled` (never left: no departure time, no arrival time), `is_diverted` (left, arrived somewhere else), and `is_disrupted` (any of cancelled, diverted, or more than 15 minutes late). Target encodings were refitted against each target — a carrier's historical *late* rate says little about its *cancellation* rate.

Recovering the 9,430 dropped flights: disruption as three outcomes



Target	Base rate	PR-AUC	ROC-AUC	Precision in riskiest 10%	Lift	Recall at 10%
Cancelled	2.3%	0.552	0.936	18.2%	8.00x	80.0%
Diverted	0.3%	0.006	0.608	0.6%	2.01x	20.1%
Any disruption	26.9%	0.566	0.732	71.1%	2.64x	26.4%

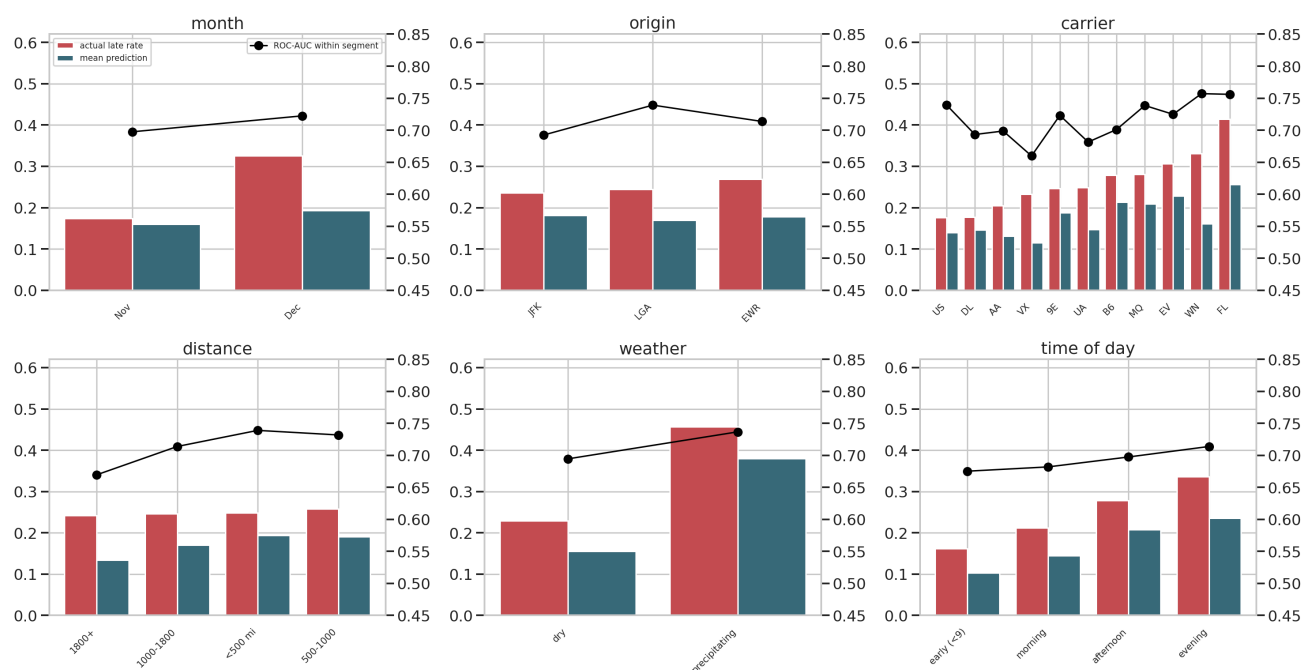
The result inverts the limitation. **The rows we threw away were the most predictable part of the problem.** Cancellation reaches ROC-AUC 0.936 — far above the 0.716 of the late/on-time task — and ranking flights by cancellation risk puts **80% of all December cancellations in the top 10% of the list**. The mechanism is the same one that makes severe delays predictable: a cancellation needs a real cause, and 8.3% of flights scheduled into a precipitating hour were cancelled against 2.0% in dry hours.

Diversions are the honest negative result: PR-AUC 0.006 against a 0.003 base rate, ROC-AUC 0.608. Barely better than chance, and it should be. A diversion is decided in the air by conditions at the *destination*, and this dataset contains weather for the three NYC origins only. There is nothing in the feature set that could predict it.

The combined disruption model is the one to deploy: PR-AUC 0.566 and 71.1% precision in the riskiest decile, both better than the late-only model, because the added outcome is the easy one.

5.8 Where the model fails

Where the model is well-calibrated and where it is not (test period, segments with $n \geq 200$)



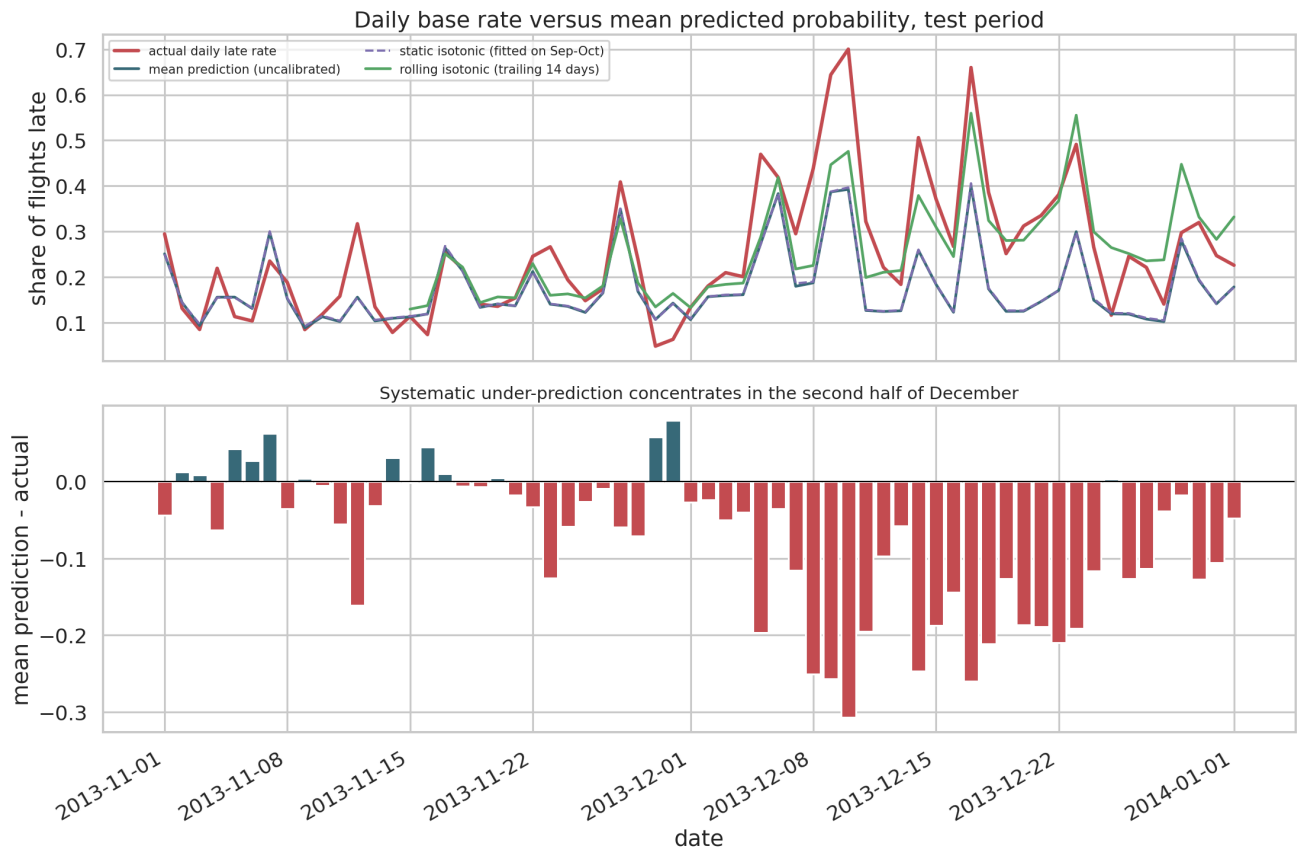
Broken out by month, origin, carrier, distance, weather and time of day (segments with $n \geq 200$):

- **Discrimination is stable.** Within-segment ROC-AUC stays between 0.66 and 0.76 everywhere. No segment is being served by noise.
- **Calibration is not.** Every segment under-predicts, and November (predicted 16.0% against 17.4% actual) is close while December (predicted 19.3% against 32.6% actual) is badly off. The failure is a systematic level shift concentrated in one month, not scattered error.
- **Long-haul is hardest.** Flights over 1,800 miles score ROC-AUC 0.670 against 0.739 for flights under 500 miles. Long flights have more time to absorb or accumulate en-route disruption that no NYC-origin feature can see.
- **Weather segments behave sensibly.** Precipitating hours score *higher* within-segment AUC (0.737) than dry hours (0.695) — when weather is doing the work, the model has something to work with.

6. Calibration under distribution shift

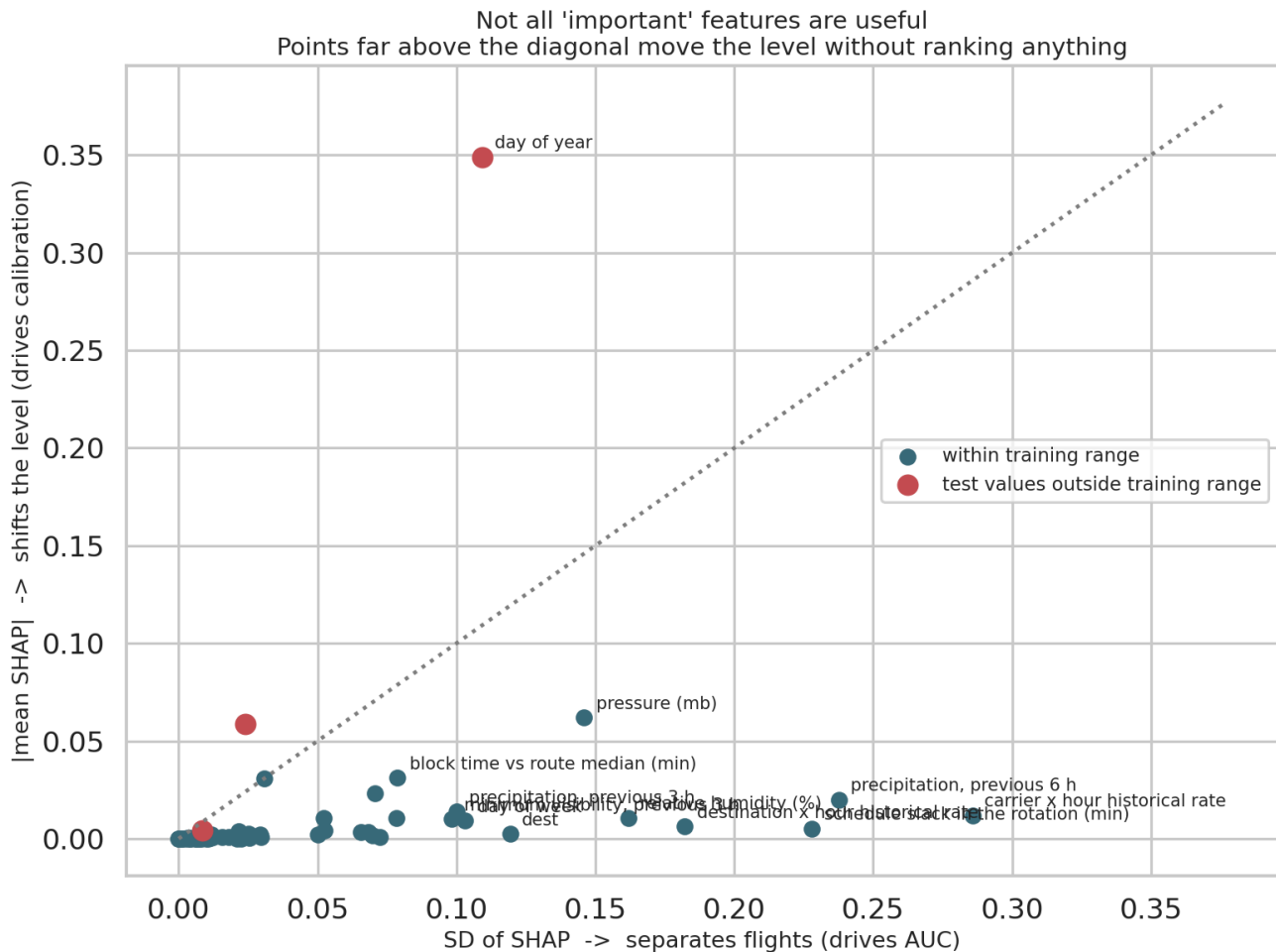
This section is the analytical core of the report.

The model's average prediction on the test period is **0.175** against an actual rate of **0.250**. It is systematically under-confident. But the failure is not uniform:



Through November the daily mean prediction tracks the daily actual rate closely, including the sharp spikes. From roughly 8 December the two diverge and never reconverge: on the worst days the model predicts 0.40 while 0.70 of flights arrive late.

6.1 Why — a diagnosis, not a guess



SHAP ranks `day_of_year` as the **single most influential feature** by mean |SHAP|, ahead of every weather and historical-rate variable. That is suspicious, and inspecting it explains the drift:

- Training values span days **1–243**. Test values span **305–365**. Every test flight is outside the range the model ever saw.
- A gradient-boosted tree cannot extrapolate. Every test flight falls into the same terminal bin on every split involving this feature.
- The result is a near-constant contribution of **−0.349 log-odds**, with standard deviation only 0.109 — Nov −0.360, Dec −0.339, essentially flat.
- The total mean SHAP across all features is −0.508, so `day_of_year` **alone accounts for 69% of the model's average downward push**.

The scatter plot above generalises this. Plotting |mean SHAP| against SD of SHAP separates two behaviours that the standard importance bar chart adds together:

- Points far **above** the diagonal shift every prediction by the same amount. They move calibration and contribute nothing to ranking. `day_of_year` is the extreme case, and all three flagged out-of-range features (`day_of_year` , `week_of_year` , `month`) sit there.
- Points **along or below** the diagonal vary flight to flight. They are what produce AUC. `te_carrier_sched_dep_hour` (SD 0.286) and `wx_precip_6h` (SD 0.236) are the real discriminators.

Mean |SHAP| conflates the two, and on a temporal split it will reliably promote calendar indices to the top of the chart while they are contributing nothing to discrimination. We have not seen this stated in the standard SHAP guidance and consider it the most transferable finding here.

6.2 The obvious fix does not work

Delete the offending features? We tested it. Removing `day_of_year`, `week_of_year` and `month` **costs 0.016 PR-AUC** (0.505 → 0.489). They earn their place through interactions with other features even though their direct contribution is a constant offset. Removing them is a net loss.

Calibrate once on the validation period? Isotonic regression fitted on Sep–Oct changes the test Brier score from 0.18006 to 0.18003 — nothing. It cannot help, because it was fitted where the base rate is 15% and applied where it is 25%, so it corrects in the wrong direction.

6.3 What does work

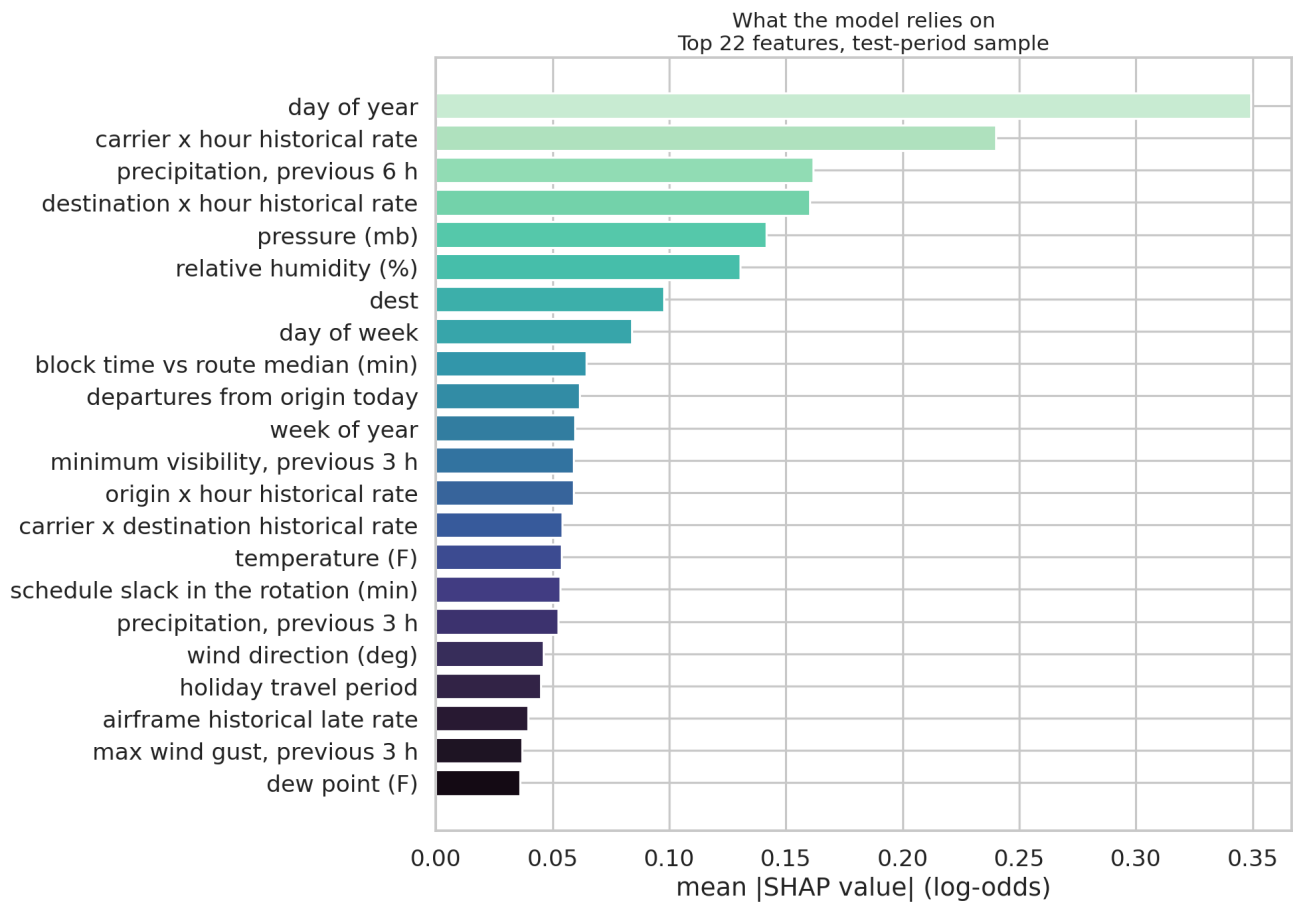
Recalibrate continuously. Each day, refit an isotonic map on the previous 14 days of flights that have already landed — labels a deployed system genuinely has — and use it to score today. Every day is scored by a calibrator that has never seen it.

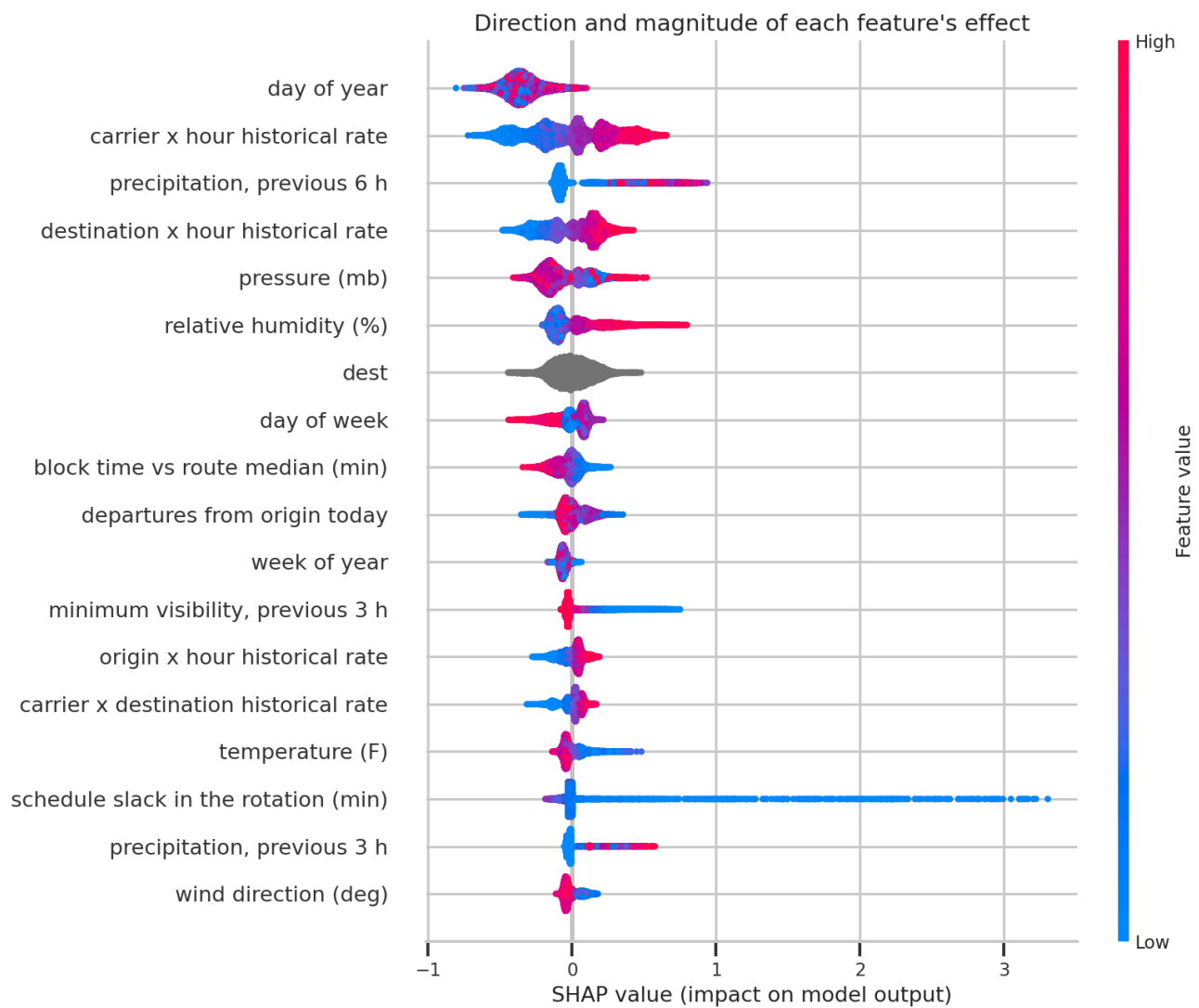
Method	Brier	Log loss	Mean prediction
Uncalibrated	0.1801	0.5526	0.183
Isotonic, fitted on validation	0.1800	0.5518	0.185
Isotonic, rolling 14 days	0.1698	0.5209	0.259
<i>(actual rate)</i>			<i>0.276</i>

A **5.7% reduction in both Brier score and log loss**, and the mean prediction moves from 0.183 to 0.259 against an actual 0.276. The green line in the drift figure tracks December where the blue and purple lines flatten.

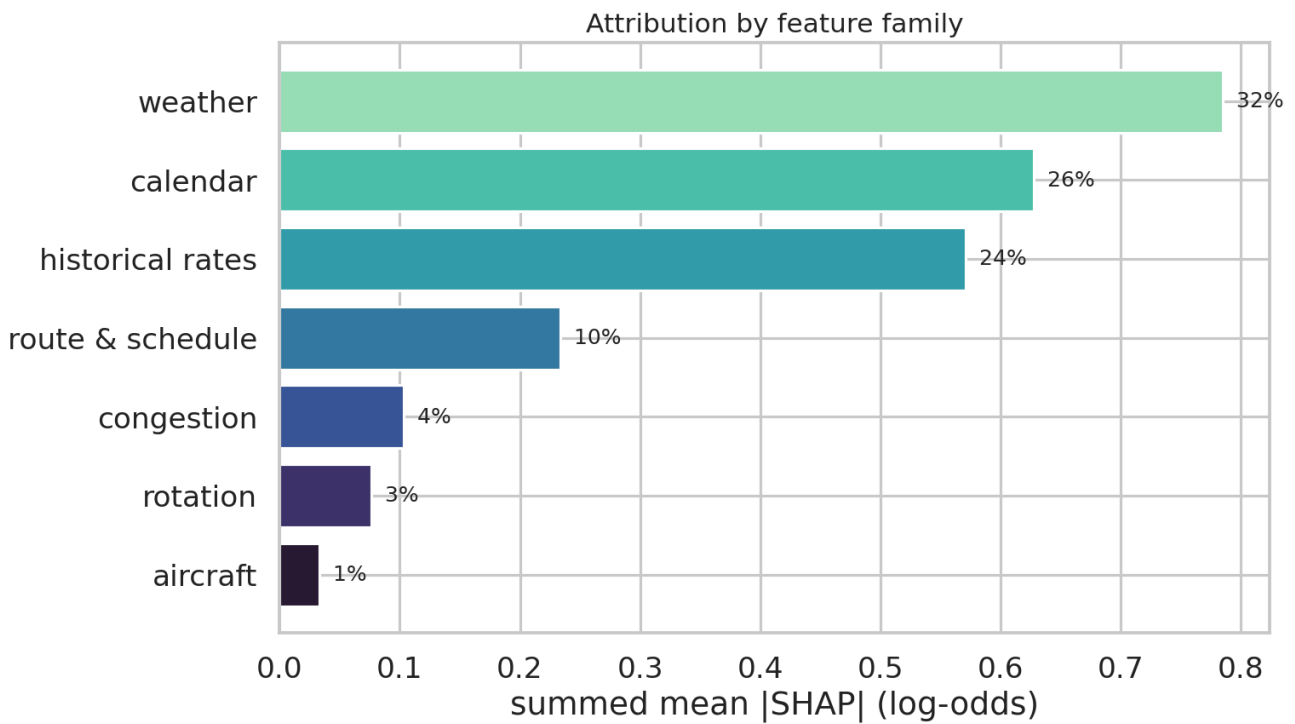
The operational conclusion: **for this problem, the maintenance burden is recalibration, not retraining.** Retraining is expensive and needs a full season of new labels; rolling recalibration needs two weeks of outcomes and a few milliseconds of isotonic regression. The discriminative structure the model learned in January still holds in December — only the level moved.

7. What the model actually uses

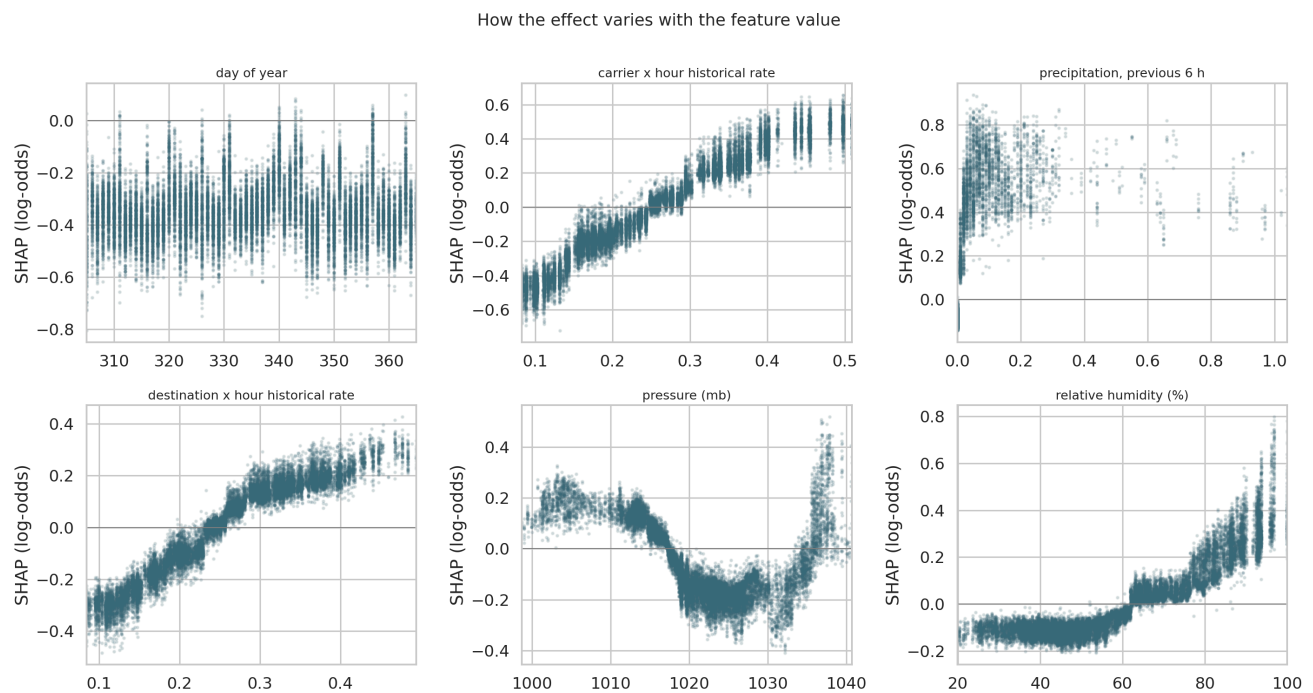




Setting aside `day_of_year` (§6.1), the top features are the carrier $\times$ hour historical rate, 6-hour precipitation, the destination $\times$ hour rate, pressure and humidity. The beeswarm confirms the directions are physically sensible: high precipitation pushes towards late, high pressure (fair weather) pushes towards on-time, high historical rates push towards late.



By family, attribution splits weather 32%, calendar 26%, historical rates 24%, route and schedule 10%, congestion 4%, rotation 3%, aircraft 1%.

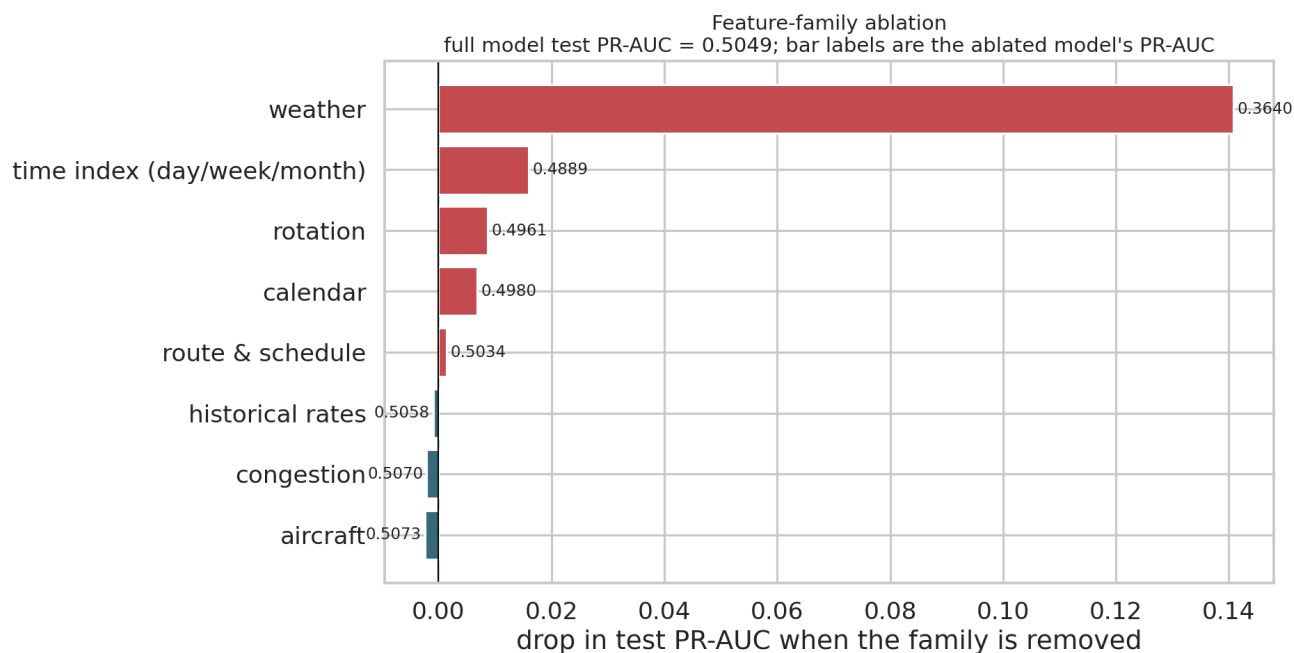


The dependence plots show threshold behaviour rather than linear response. Precipitation has almost no effect until it becomes non-zero, then jumps. Visibility matters below about 5 miles and is flat above it. This is why tree ensembles beat logistic regression here, and also why the margin is only 0.029 — the thresholds are sharp but few.

7.1 Attribution is not value: the ablation

SHAP measures how much the *fitted model leans on* a feature. That is not the same as how much predictive value the feature *carries*, because a feature can be heavily used and entirely redundant. To measure value we

retrained the model with each family removed and recorded the drop in test PR-AUC.



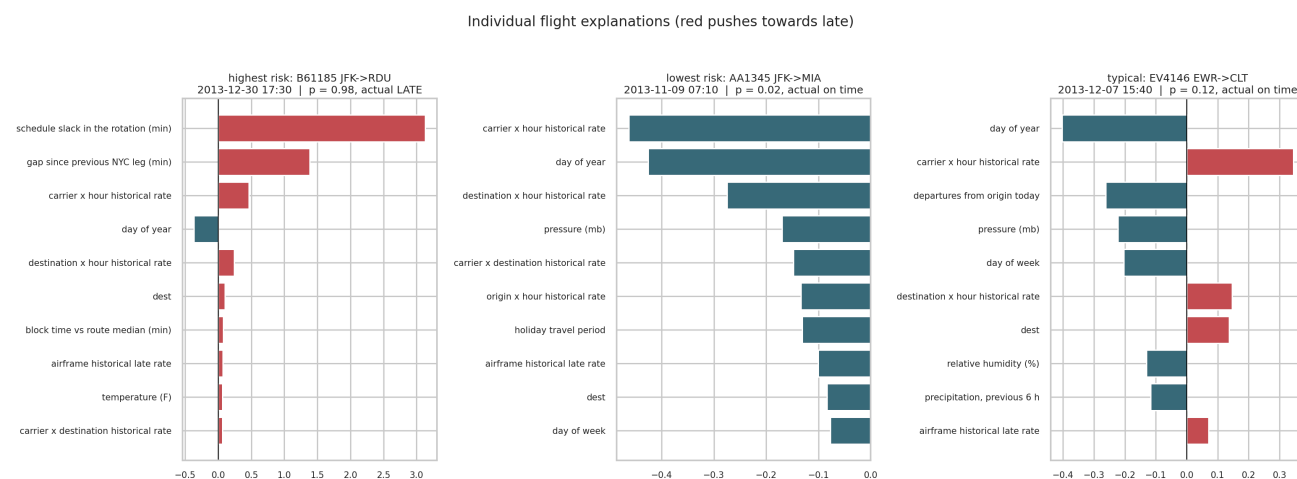
Family removed	Test PR-AUC	Change
— (full model)	0.5049	—
Weather	0.3640	-0.1409
Time index (day/week/month)	0.4889	-0.0160
Rotation	0.4961	-0.0088
Calendar (all)	0.4980	-0.0069
Route & schedule	0.5034	-0.0015
Historical rates	0.5058	+0.0009
Congestion	0.5070	+0.0021
Aircraft	0.5073	+0.0024

Three conclusions, two of them uncomfortable:

- 1. Weather is the entire game.** Removing it costs 0.141 PR-AUC — ten times the next-largest effect, and it drops the model below logistic regression *with* weather. If this system were being productionised, the hourly weather feed is the one input worth paying for, and a weather **forecast** feed would be the highest-value extension.
- 2. The historical-rate encodings are redundant.** Eight target encodings, five-fold out-of-fold machinery, Bayesian smoothing — and removing them all *improves* test PR-AUC by 0.0009. SHAP attributes 24% of influence to them, yet whatever they encode is already recoverable from carrier, destination and hour as raw features. This is exactly the attribution-versus-value gap.
- 3. Congestion and aircraft features are noise.** Both marginally negative. Scheduled congestion is presumably already implicit in hour-of-day, and aircraft age tells you about the airline's fleet, not about today.

A leaner model — weather, calendar, route, carrier, rotation — would score within noise of the full one at roughly a third of the features. We report the full model because it is what the search selected, but the ablation is the more useful result for anyone building on this.

7.2 Individual explanations



Because SHAP decomposes each prediction exactly, every score comes with a reason an operations agent can read. The highest-risk flight in the test sample — B6 1185, JFK→RDU, 30 December at 17:30, $p = 0.98$, which did arrive late — is driven almost entirely by rotation slack (+3.1 log-odds): the timetable gave that airframe an impossible turnaround on one of the worst days of the year. The lowest-risk flight, AA 1345 JFK→MIA at 07:10 on 9 November, is pushed down by the carrier × hour rate and the early departure slot.

8. Limitations

Stated plainly, because most of them bound the result.

Three of the limitations in the first version of this report have since been measured rather than asserted, and are recorded here as resolved:

- ~~Cancellations and diversions are dropped.~~ **Resolved in §5.7.** All 336,776 flights are now modelled as three outcomes. Cancellation turned out to be the most predictable outcome in the study (ROC-AUC 0.936). Diversion remains genuinely unpredictable from NYC-origin features, which is itself the answer.
- ~~Weather is observed, not forecast, and we have not measured the cost.~~ **Resolved in §5.6.** A persistence-forecast horizon curve puts a lower bound on it: three hours costs 0.020 PR-AUC, six hours retains 72% of weather's contribution, twenty-four hours retains none.
- ~~The severity head barely beats a constant.~~ **Resolved in §5.5.** Reframed as tiers and quantiles, where the same features support ROC-AUC 0.793 on the >120-minute tier and a P90 head 16.4% better than a constant.

What remains:

1. **Diversions are not predictable here** (§5.7), and no amount of feature work on NYC-origin data will change that. It needs destination weather and en-route conditions.

2. **The horizon curve is persistence, not forecast.** It bounds the answer from below, which is the safe direction, but a real NWP forecast feed would land somewhere above the curve and we cannot say where without one.
 3. **The inbound leg is invisible.** Only NYC departures are recorded, so the dominant propagation mechanism is only partly observable (§3.3). A full BTS extract with all US flights would let the true inbound aircraft be tracked and is the single most promising data extension.
 4. **One year, three airports.** No evidence the model transfers to other airports or years. 2013 predates several changes in FAA slot rules and airline scheduling practice.
 5. **Calibration degrades under seasonal shift** and requires rolling recalibration (§6). A deployment without it will systematically under-warn in the worst weeks — precisely when it matters most.
 6. **The 4:1 cost ratio is asserted, not derived.** The threshold moves with it. We swept the trade-off (figure 10) so a different ratio can be read off, but a real deployment needs the airline's actual numbers.
 7. **Congestion counts use the full-year timetable.** Legitimate, since the schedule is published in advance — but a strict production replay would freeze the timetable as known at prediction time, which differs slightly after cancellations.
-

9. What the model is worth

Every number so far measures the ranking. This section measures the consequence of using it, which is a different question and needs a different kind of honesty: the chain from a probability to a dollar has three links, and only the first belongs to us.

1. **Ranking quality.** Measured, on held-out data, above.
2. **Unit costs.** External and cited: \$98.41 per block minute, \$47 per passenger hour, an 83.1% load factor (§1.1).
3. **Mitigation effectiveness** — how much of a warned delay a desk actually recovers. **Nobody publishes this, and we cannot measure it from a historical dataset**, because the dataset contains no counterfactual. It is therefore never assumed: the headline uses a deliberately pessimistic 10%, the full 2–40% range is swept, and the break-even is reported so the reader can substitute their own belief.

9.1 The operating rule, made harder on purpose

The budget is spent **per day**, not pooled across the test period. This matters more than it sounds. A desk does not get to save November's alerts and spend them all on 22 December; pooling would let the evaluation do exactly that and would flatter the result. Under the pooled rule the top decile is 64% late (§5.3). Under the per-day rule it is **47.2%**. The lower number is the one this section uses.

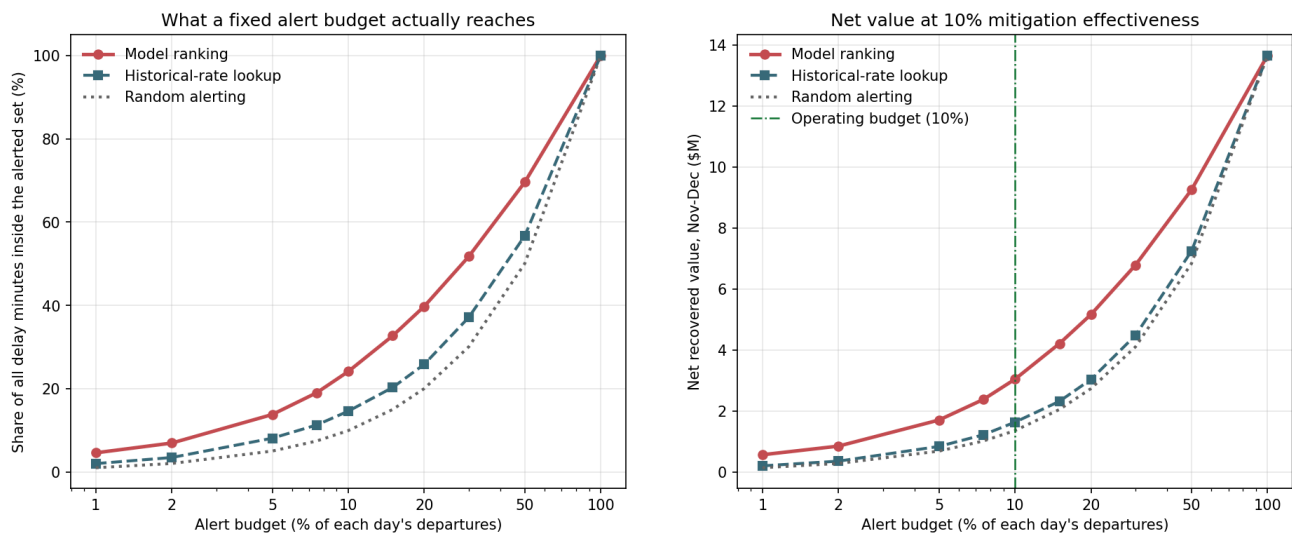
9.2 What a 10% budget reaches

Nov–Dec 2013: 53,991 flights, 13,492 late, **761,202 delay minutes**, 1,381,641 passenger-hours of delay.

Rule	Alerts	Precision	Delay minutes reached	Share of all delay
Random	5,423	25.0%	75,965	10.0% $\pm$ 0.3%
Route's historical late rate (no ML)	5,423	31.6%	111,426	14.6%
This model	5,423	47.2%	183,879	24.2%

The comparison that matters is the first row, not zero. A desk alerting at random on 10% of flights already catches 10% of the delay minutes; a claim of value has to be stated net of that. The model reaches **2.4× the delay minutes random selection does** and **1.65× a route-level historical lookup table** — the thing an airline would plausibly build in a spreadsheet instead.

Operational impact on the held-out Nov-Dec 2013 period



Note the shape of the left panel: the model's advantage is widest at small budgets and vanishes at 100%, where every rule alerts on everything and ranking stops mattering. That is the correct behaviour and a useful sanity check — a curve that did not converge would mean a bug.

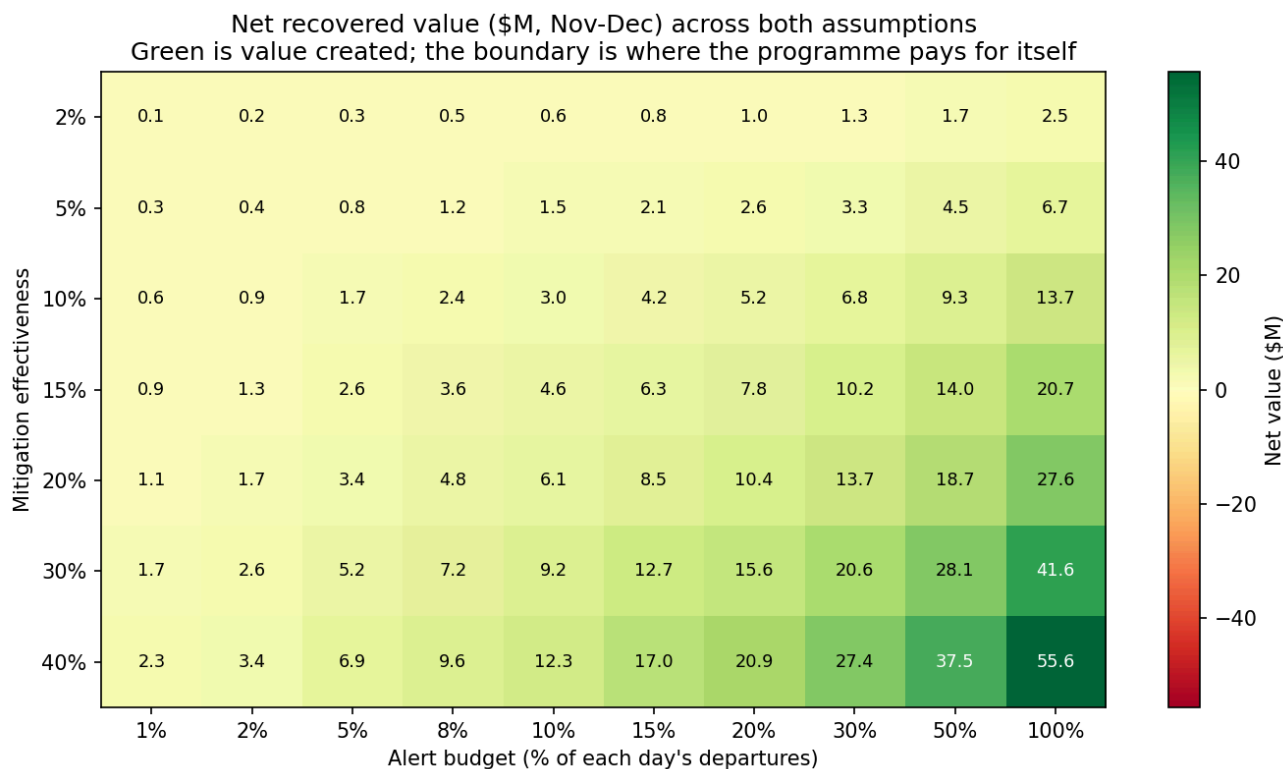
9.3 Turning that into value

At 10% mitigation effectiveness, over the two held-out months:

Quantity	Value
Delay minutes recovered	18,388
Airline operating cost avoided	\$1.81M
Passenger time returned	27,015 hours (\$1.27M)
Desk cost of handling 5,423 alerts	\$0.03M
Net recovered value	\$3.05M
— of which attributable to the model rather than random alerting	\$1.69M
CO ₂ not burned	331 tonnes

Scaled to a full year of New York departures: **\$18.0M–\$19.0M**. The range is not decoration. November and December run late more often than the year as a whole (25.0% against 23.7%), so the naive annualisation is an upper bound and the seasonally discounted figure is the lower one. We do not claim to know where in that band the truth sits. We also do not extrapolate to the national network: multiplying a three-airport result by the size of the US system would be arithmetic, not evidence.

9.4 Everything above, swept



Break-even mitigation effectiveness is **0.11%** — and that number is the finding, not a victory lap. Handling an alert costs about \$6; the delay it targets is worth about \$568. The system is nowhere near being gated on cost. **The binding question is whether advance warning changes an outcome at all**, and this dataset cannot answer that, because it records what happened and not what would have happened.

One-at-a-time on the external constants, net value stays positive throughout:

Constant varied	Low	High	Swing
Block-minute cost (\$70 / \$130)	\$2.52M	\$3.63M	\$1.10M
Passenger value of time (\$30 / \$70)	\$2.59M	\$3.67M	\$1.08M
Desk cost per alert (\$2 / \$20)	\$3.07M	\$2.97M	\$0.10M

The conclusion does not depend on any single published figure. It depends entirely on the one figure nobody publishes, which is stated plainly rather than hidden inside an average.

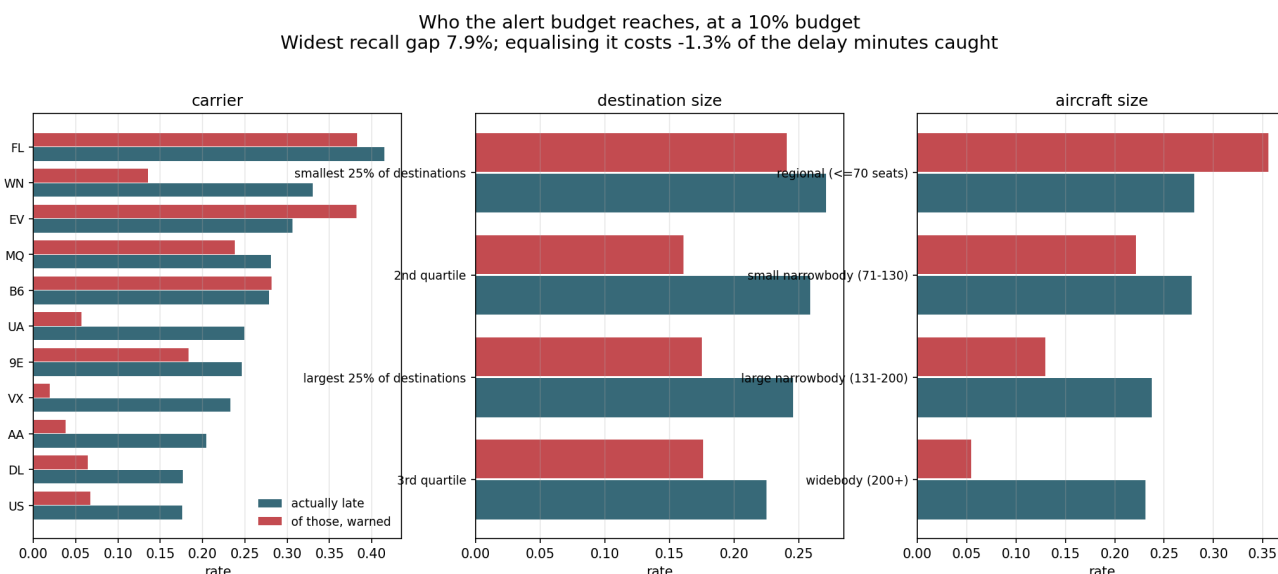
9.5 What this section is not

It is not a business case, and it is not a measured saving. It is a ceiling: the value *available* to a desk that acts perfectly on the list, times an assumed recovery rate. A real evaluation needs a controlled rollout — alert on a

random half of the eligible flights and compare — and that is the first thing we would build if this went past a hackathon.

10. Who the alerts reach

A ranking model with a fixed budget is a rationing device, and "highest probability first" is not a neutral rule. It gives the budget to whichever groups already run late most often, which is efficient and is not the same as fair. `src/fairness.py` measures where the alerts land instead of assuming it is fine.



10.1 Coverage is very uneven across carriers

At a 10% daily budget, the share of a carrier's genuinely-late flights that got warned about — the equal-opportunity criterion — ranges from **38.2% to 1.9%**, a **36.3-point gap**:

Carrier	Flights	Ran late	Of those, warned	Share of alerts	Calibration error
FL	410	41.5%	38.2%	2.3%	-0.158
EV	8,272	30.6%	38.2%	36.3%	-0.078
B6	8,929	27.8%	28.2%	30.4%	-0.065
WN	2,100	33.0%	13.5%	2.5%	-0.170
UA	9,646	24.9%	5.7%	5.3%	-0.102
AA	5,147	20.4%	3.8%	1.7%	-0.073
VX	902	23.3%	1.9%	0.1%	-0.118

Southwest is the clearest case of the failure mode. It runs late on a third of its flights — a higher rate than JetBlue — and receives 2.5% of the alerts, because the model under-predicts it by 17 points. Its alerts are unusually *accurate* when they fire (68.6% precision); there are simply far too few of them. That is not a ranking problem the AUC would ever show.

Every group has a negative calibration error, which is the December under-prediction of \$6 reappearing: the model is not miscalibrated *unevenly* so much as miscalibrated *everywhere*, with the size of the miss varying by carrier.

10.2 Pricing the trade-off instead of naming it

The usual move here is to note the disparity and move on. Instead we re-allocated the **same** daily budget so each group receives alerts in proportion to its share of that day's departures, ranking by risk within each group. Largest-remainder rounding keeps the alert count identical (5,423), and the allocation stays per-day, so this is not a pooled rule getting an unfair advantage.

Grouping	Coverage gap, global rule	...proportional	Delay minutes reached	Change
Carrier	36.3%	6.4%	183,879 → 162,630	-11.6%
Destination size	7.9%	1.4%	183,879 → 186,269	+1.3%

Two different answers, and the second is the interesting one. Equalising across carriers costs 11.6% of the delay minutes caught — a real trade-off, now priced rather than argued about. Equalising across destination-size quartiles **gains** 1.3%: for that grouping, fairness is free. That is not a free lunch, it is a symptom — it means the global threshold was mis-allocating across those groups because the model's calibration differs between them, and evening out the budget partially corrects the error.

The recommendation that follows is concrete: **allocate the alert budget per-carrier rather than globally**, accept the 11.6% efficiency cost, and treat the residual gap as a monitored metric rather than a solved problem.

10.3 What we did not check

Passenger-level protected attributes are not in this dataset and cannot be inferred from it responsibly, so nothing here is a claim about demographic fairness. The groups audited are operational — carrier, destination size, aircraft size, time of day, origin airport — and destination size is a proxy for passenger slack, not for any characteristic of the people on board.

11. Running this after the hackathon

11.1 What it costs to operate

The deployable classifier is a 5.2 MB LightGBM booster; the full set of heads (severity, quantiles, tiers, cancellation) comes to 64 MB on disk, and a production deployment needs the first one plus whichever heads it displays. A day of New York departures is roughly 900 rows of tabular data. There is no GPU, no vector database and no inference API bill. A realistic production footprint:

Component	Requirement	Notes
Nightly retrain	~10 min, 4 cores	not needed nightly — see §6, recalibration beats retraining
Rolling recalibration	seconds, daily	the piece that actually needs to run daily
Scoring	one batch of ~900 rows per day	a cron job, not a service
Weather feed	hourly METAR/ASOS	free from NOAA; already the input format
Schedule feed	the airline's own timetable	already exists in every ops system
Storage	<100 MB/year of predictions	

The only genuinely recurring cost is a person to look at the list, which is the cost the impact model already charges against the benefit (\$6 per alert, swept to \$20 without changing the sign).

11.2 Who maintains it

The maintenance burden is small but not zero, and the honest version is that it concentrates in one place: **calibration drift**. §6 showed that the model's discrimination survives the seasonal shift while its calibration does not, and that a rolling 14-day isotonic recalibration fixes it where retraining does not. So the operating procedure is a daily recalibration job on the previous fortnight of landed flights, plus the monitoring in §10.1 as a standing report. `make verify` exists precisely so that whoever inherits this can tell in one command whether it still does what the report says.

11.3 Environmental and social effects, both directions

Environmental. Recovered delay minutes are unburned fuel. At roughly 18 kg CO₂ per delay minute — derived from a narrowbody's idle burn, halved, because delay-minute burn runs about half the unimpeded taxi rate⁴ — a 10% budget at 10% effectiveness avoids **331 tonnes over two months**, or roughly 2,000 tonnes a year across the three airports. That figure is order-of-magnitude and labelled as such; it is swept from 9 to 27 kg/minute in `impact.json`. Against it: training the models once costs a few core-hours, which is negligible on any comparison.

Social, and the risk that runs the other way. A delay-risk score is a ranking of flights, and rankings get reused for things they were not validated for. Three failure modes are foreseeable and worth naming before somebody discovers them:

- *Feedback.* If a carrier is systematically under-alerted (§10.1), its delays go unmitigated, its historical late rate rises, and a retrained model learns the higher rate. Per-carrier budget allocation breaks that loop; global thresholding does not.
- *Misuse against passengers.* Nothing here should touch an individual passenger — not boarding, not pricing, not compensation eligibility. The model card in the app says so explicitly.
- *False confidence.* A 47% precision list is a triage aid. Presented as a prediction, it will be wrong slightly more often than it is right, and the app's default view is built around that framing on purpose.

11.4 A credible path past the demo

The next step is not a bigger model. It is the one measurement this dataset cannot supply:

1. **Swap the persistence weather for a real forecast feed.** §5.6 shows a three-hour horizon costs 0.020 PR-AUC using the crudest possible forecast, so the current numbers are a floor. This is the cheapest

available gain.

2. **Run a randomised rollout.** Alert on a random half of eligible flights and compare outcomes. That measures mitigation effectiveness directly and replaces the one assumption §9 has to sweep.
3. **Extend beyond three origins.** The BTS on-time database has the same schema for every US airport; the feature code is origin-agnostic, and the inbound-leg blind spot in §8 closes as soon as the network is complete rather than NYC-only.
4. **Model three outcomes, not two.** §5.7 already shows cancellation is the most predictable outcome in the dataset (ROC-AUC 0.936). A production system should rank on-time / late / cancelled jointly.

Steps 1 and 4 are code we could write with this dataset. Step 2 is the one that would turn §9 from a ceiling into a measurement, and it requires an airline.

12. Conclusions

On the problem. Arrival delay is predictable in advance to a useful but bounded degree: PR-AUC 0.507 against a 0.250 base rate, with 64% precision on the riskiest 10% of flights. That is a genuine triage tool and not a crystal ball. The much higher numbers commonly reported for this dataset come from including the observed departure delay, which lifts PR-AUC to 0.846 and answers a question nobody needs answered.

On what matters. Weather dominates everything else by an order of magnitude — and a persistence forecast retains 86% of that value three hours out, so the system is deployable well before push-back rather than only at the gate. Feature construction beat model selection: logistic regression on good features beat the historical-rate rule by 0.138 PR-AUC, while gradient boosting on those same features beat logistic regression by only 0.029. XGBoost finishes 0.006 ahead of LightGBM once both receive 40 draws and the same categorical representation — a gap that only appeared after the comparison was deliberately re-levelled, which says more about how library benchmarks are usually run than about either library.

On what is predictable. The same pattern turned up in three independent experiments: **the worse the outcome, the better it is predicted.** ROC-AUC rises from 0.716 on the >15-minute tier to 0.793 on >120 minutes. The P90 quantile head beats a constant by 16.4% where the P50 head manages 2.0%. And cancellation — the outcome originally discarded as unlabellable — reaches 0.936, putting 80% of December's cancellations in the top decile of risk. Severe disruption has causes that are in the feature set; marginal lateness is mostly noise. A model built for this problem should be aimed at the tail, and the first version of our severity head failed precisely because it was aimed at the middle.

On method. Three of our findings are about method rather than flights, and they are the ones most likely to transfer:

- A target-encoding key that does not recur across a temporal split is worse than no feature at all (§3.2).
- Mean |SHAP| conflates level shifts with ranking power, and on temporal splits it will promote out-of-range calendar features to the top of the importance chart while they contribute nothing to AUC. Plotting mean against standard deviation separates them (§6.1).
- Feature attribution and feature value diverge sharply. The historical-rate encodings took 24% of SHAP attribution and were worth -0.0009 PR-AUC (§7.1). Only an ablation answers the question "is this data worth collecting".

On what it is worth. A 10% daily alert budget reaches 24.2% of all delay minutes, against 10.0% for random alerting and 14.6% for a route-level historical lookup — 2.4× and 1.65× respectively. At a pessimistic 10% mitigation effectiveness that is \$3.05M of recovered value over two months, \$1.69M of it attributable to the model rather than to the budget existing at all. Break-even sits at 0.11% effectiveness, which means cost is not what gates this system; the open question is whether advance warning changes an outcome at all, and no historical dataset can answer it.

On who it serves. Ranking purely by probability is a rationing rule with winners and losers: coverage of genuinely-late flights ranges from 38.2% to 1.9% across carriers. Southwest runs late more often than JetBlue and receives a fortieth of the alerts, because the model under-predicts it by 17 points — a failure invisible in the AUC. Allocating the same budget proportionally cuts that gap from 36.3 to 6.4 points for 11.6% of the delay minutes caught, and for destination-size quartiles it *gains* 1.3%. That is the recommendation: budget per carrier, not globally.

On deployment. Ship the pre-flight model with a cost-derived threshold of 0.20 rather than 0.5 — at the default the model costs 31% more per flight than at the tuned threshold, and more than alerting on every flight — and a rolling 14-day isotonic recalibration, which recovers 5.7% of Brier score under the December shift that retraining alone would not catch. Allocate the alert budget per carrier, and monitor the coverage gap as a standing metric rather than treating it as solved.

Appendix A — Reproducing this report

```
pip install -r requirements.txt
make reproduce          # ~15 minutes on 4 cores
```

Every figure and every number above is regenerated by that command. Seeds are fixed (`SEED = 42` in `src/config.py`); random search draws are seeded, so draw *i* is always the same hyperparameter set.

Appendix B — Artefacts

File	Contents
reports/metrics/evaluation.json	every test metric, threshold sweep, calibration comparison
reports/metrics/horizon.json	forecast-horizon curve, 7 horizons
reports/metrics/disruption.json	cancellation / diversion / disruption models
reports/metrics/severity_v2.json	tiers, quantile heads, conditional-on-late
reports/metrics/model_fingerprints.json	booster SHA-256 checksums
reports/metrics/lgbm_search.json	all 40 search draws with per-fold scores
reports/metrics/xgb_search.json	8 XGBoost draws
reports/metrics/ablation.json	feature-family ablation
reports/metrics/shap_importance.json	SHAP importance, group shares, out-of-range diagnostics
reports/metrics/segment_errors.csv	per-segment error analysis
reports/metrics/threshold_sweep_validation.csv	full precision/recall/cost curve
reports/metrics/eda_summary.json	descriptive statistics
reports/metrics/impact.json	delay minutes, passenger hours, dollars and CO ₂ at every budget, with all assumptions swept
reports/metrics/impact_budget_curve.csv	the budget curve, model vs historical rule vs random
reports/metrics/fairness.json	per-group coverage audit and the priced equity trade-off
reports/metrics/fairness_*.csv	one table per grouping
reports/figures/01–29	all figures

Appendix C — Test suite

16 checks in `tests/test_features.py`, run with `make test`:

Leakage — pre-flight features exclude all outcome columns; gate mode adds exactly the three post-push-back features; the inbound-leg delay never reads from the future; the tail-number encoding is genuinely out-of-fold; every encoding key recurs across the split boundary.

Split integrity — splits are disjoint, ordered in time, and no rows are lost across the joins.

Feature correctness — timezone-corrected block time matches air time plus plausible taxi and JFK–LAX comes out near six hours; HH:MM conversion; congestion counts nest correctly; rotation gaps are non-negative and same-day; the target matches its definition; the weather join is complete after filling; the weather join key is unique (the DST trap of §2.2); every forecast horizon joins one-to-one; no object dtypes reach the model.

1. Ball et al., *Total Delay Impact Study*, NEXTOR for the FAA; figures as reported by Airlines for America and widely cited at \$33bn/year, of which \$16.7bn falls on passengers and \$8.3bn directly on carriers. ↩
2. Airlines for America, "U.S. Passenger Carrier Delay Costs", 2025 calendar year, published 23 July 2026, from DOT Form 41 filings. <https://www.airlines.org/dataset/u-s-passenger-carrier-delay-costs/> ↩
3. FAA, *Economic Values for FAA Investment and Regulatory Decisions*, Section 1: value of travel time, cited by A4A as the \$47/hour figure. ↩
4. Ryerson, Hansen & Bonn, "Estimating fuel burn impacts of taxi-out delay with implications for gate-hold benefits", *Transportation Research Part C*, 2016 — fuel consumption per minute of taxi *delay* is roughly half the unimpeded taxi rate, so applying nominal taxi burn to delay minutes overstates it. Combined with 9.57 kg CO₂ per gallon of jet fuel (EPA). ↩